Gaze Sampling of Motion Relates Differently to Authenticity Judgments for Real and AI-Generated Physics Videos

ANONYMOUS AUTHOR(S)

This paper investigates the relationship between motion observed during fixation and human judgments regarding the authenticity of AI-generated videos. The analysis involved a duration-weighted coupling score that compared motion at fixations with same video, time, eccentricity-matched controls on 385 judgment trials from 40 participants. Under this primary reference, higher coupling accompanied fewer AI judgments for real clips and more for AI clips (ratio of odds ratios = 3.12, 95% CI [1.54, 6.33]). Adding coupling and its class interaction improved the fit of a logistic mixed model of AI versus Real responses containing actual class, mean motion, and conventional gaze summaries. Associations for AI clips were imprecise when controls incorporated gaze history or image content. These findings link authenticity judgments to motion sampled during viewing, while leaving open whether motion informed decisions, developing judgments guided gaze, or both.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Computer vision.*

Additional Key Words and Phrases: eye tracking, dynamic scene viewing, generative video, visual attention, gaze–motion coupling, persistent homology

1 Introduction

AI-generated videos of physical events raise distinct questions about visual appearance, temporal structure, and physical consistency. Benchmark work on generative video has treated physical principles as an evaluation target separate from visual realism, motivating measurements that describe temporal image structure without treating those measurements as direct ratings of plausibility [20]. Viewers, meanwhile, make authenticity judgments while sampling only part of a changing scene. Eye tracking can characterize where and when that sampling occurs, but a final real-or-AI response does not by itself identify the visual information encountered during viewing.

Motion is a cue viewers themselves report when assessing generated video [7]. However, retrospective explanations need not identify all information contributing to a decision[21], and verbal reports require interpretation in light of how and when they are elicited[10]. Eye tracking permits a complementary question: did a viewer sample more motion than same-time, location-matched alternatives, and did that sampling relate to the judgment?

Authenticity judgments may depend on the motion sampled at particular locations and moments, whereas scanpath length, dispersion, and entropy summarize viewing more broadly. We reanalyze the physics-video subset of Shi et al.'s eye-tracking study [26]: 20 matched real–AI scene pairs and 385 eligible T2 judgment trials from 40 participants. The main question is whether controlled gaze sampling of motion relates to the participant's Real/AI response differently for actual-real and actual-AI clips. Actual clip class is supplied to these response models; their target is the viewer's judgment conditional on known classes.

The open research question is whether motion sampled at fixation, relative to motion available at the same video times and eccentricities, relates to authenticity judgments, and whether this relationship differs between real and

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

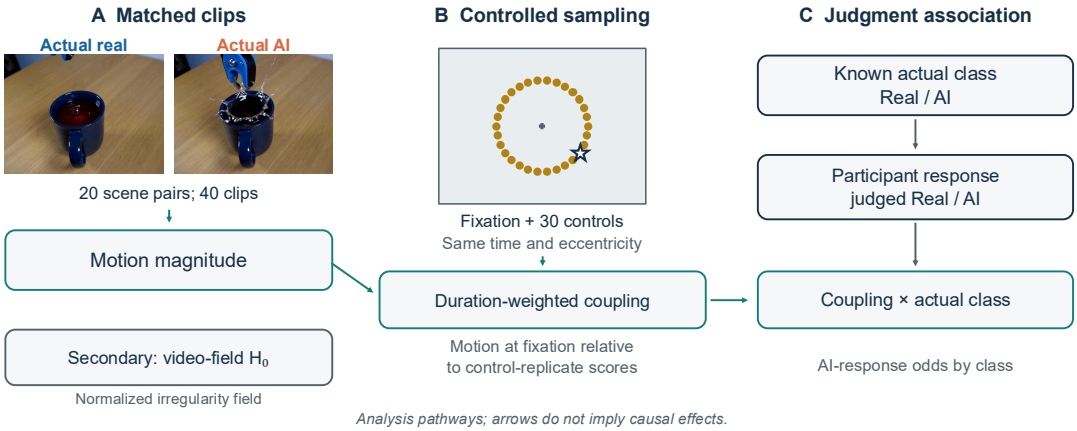


Fig. 1. **Study overview.** Matched real and AI physics clips supply motion fields. Controlled coupling compares motion sampled at fixation with 30 same-time, eccentricity-matched controls, retaining fixation timing and duration. The judgment model distinguishes known actual clip class from the participant's Real/AI response. Video-field persistence and other exploratory comparisons provide secondary descriptions. Arrows mark analysis pathways, not causal effects.

AI-generated clips. Addressing this question connects gaze to the motion available at the particular locations and moments viewed, which conventional gaze summaries and retrospective cue reports do not directly capture. The contribution of this paper is a controlled assessment of how moment-specific motion sampling relates to authenticity judgments. We quantify this sampling with a duration-weighted gaze–motion coupling measure. The result of the analysis shows that, under the primary eccentricity-matched reference, greater coupling relates to lower odds of an AI judgment for real clips and higher odds for AI clips. Coupling also improves judgment-model fit beyond conventional gaze summaries. Exploratory reference, video-structure, and stated-cue analyses examine the scope and limitations.

2 Related Work

2.1 Motion, task, and gaze allocation in dynamic scenes

Dynamic visual features predicted saccade targets particularly well soon after scene onsets in the edited-video experiments of Carmi and Itti [5]. That result provides a basis for measuring motion sampled at gaze locations, while the scene and task determine how such sampling should be interpreted. Motion is one feature that can organize gaze during dynamic-scene viewing, although its influence depends on the scene and task [19]. During viewing of unedited real-world videos, task instructions shifted gaze away from high-flicker regions and people after initially center- and motion-biased saccades [27]. In a dynamic body-expression task, gaze allocation varied with the informativeness of facial and movement cues [23]. Gaze distributions in natural dynamic scenes can also reflect center-related and composition-related biases [30]. A raw average of motion at fixation locations may therefore conflate motion sampling with where viewers tend to look. Clarke et al. [6] additionally motivate a reference conditional on the preceding fixation. We use a within-video control that matches gaze locations by radial eccentricity and samples them at the same moments as observed fixations. The resulting score quantifies the magnitude of relative motion at sampled gaze positions. Dynamic scenes often stimulate smooth pursuit of moving objects, making it challenging to distinguish these movements from fixations. This difficulty has led to the development of specialized detection methods that aggregate

gaze data from multiple observers [1]. Since the current data consist of fixation records, the score indicates where fixations occurred relative to motion, rather than confirming whether the eyes actively tracked the movement.

2.2 Gaze and evaluative judgments

Across 23 comparative studies, a review found no modality-independent gaze signature of AI-generated content and called for clearer links between gaze measures and functional outcomes [24]. In eye-tracking research, gaze has been examined in relation to authenticity in two ways. First, as a characteristic of the stimulus, inconsistencies in the eye and gaze behavior of synthetic faces have been utilized to identify deepfake videos [8]. Second, as an indicator of the viewer's behavior, studies have shown that radiologists exhibit different fixation and saccade patterns when viewing real versus AI-generated chest radiographs [31]. This paper falls into the second category, focusing on dynamic physics scenes, non-expert viewers, and motion captured at specific moments, rather than on static images. Authenticity responses also need to be distinguished from ground truth: participants can judge real face photographs as AI-generated [18]. In video viewing without eye tracking, longer viewing improved generated-clip detection and participants reported motion among their cues [7]. That study collected strategy reports at the end of the survey, leaving the correspondence between stated cues and motion sampled by gaze on individual trials unresolved by its design. In a different eye-tracking task, Papadopoulos et al. linked gaze features to preference and confidence during pairwise AI-image evaluation [22].

Verbal explanations and online sampling address related but distinct questions. Nisbett and Wilson [21] described limits on reporting the processes underlying decisions. Post-decision explanations can also be given for choices whose outcomes were covertly reversed [15]; such reports are therefore treated here as stated cue content. Shi et al. [26] recorded gaze in video-understanding (T1) and AI-detection (T2) tasks. Broad gaze measures did not differ reliably by actual authenticity, while task and perceived authenticity related to viewing behavior. Our analysis reuses that cohort and selects only physics clips. It adds a same-time, eccentricity-controlled motion measure and asks about conditional responses.

2.3 Between-viewer agreement and topological summaries

Dorr et al. [9] showed that between-viewer gaze variability depends on dynamic scene content and viewing conditions. Gaze overlap among observers watching the same video has been quantified from their timestamped gaze paths [29]. In video-lecture learning, Bühler et al. [3] found greater gaze synchrony during self-reported attention, while the synchrony measures did not correlate with learning outcomes. Agreement across viewers and an individual's task response are therefore distinct quantities. Our exploratory diagram-distance measure compares space-time gaze-field topology rather than simultaneous fixation overlap.

We also summarize temporal structure with zero-dimensional persistent homology. Across a sequence of thresholds, connected regions appear and merge; their birth and death values provide a multiscale description of connectivity [4]. Here, the calculation is applied to a three-dimensional, per-clip normalized field of forward–backward flow and photometric inconsistency, and H_0 total persistence sums finite connected-component lifetimes. Note that it is an algorithmic descriptor of the processed field, not a semantic map of artifacts or violations of physical laws. Prior topological work on gaze has addressed raw trajectory segmentation[13] and, more recently, fixation sequences treated as one-dimensional coordinate time series, whose persistence features complemented conventional features for dyslexia detection [14]. Our gaze measure instead applies cubical persistence to a smoothed, duration-weighted space–time gaze density field[28] and is used descriptively rather than as a classification feature.

3 Methodology

3.1 Participants, stimulus subset, and procedure

We reanalyzed Shi et al.'s eye-tracking study [26], selecting its physics set (S1): 40 adults with normal or corrected-to-normal vision (26 women; ages 18–48). The source study had ethics approval and informed consent; this work used de-identified records without new recruitment. S1 comprised 20 real Physics-IQ videos [20] and 20 AI clips generated with Google Veo 3.1 (October 2025) from text descriptions and keyframes of the corresponding scenarios. Scenes covered solid mechanics, fluids, optics, thermodynamics, and magnetism. Clips were cropped to 960×720, trimmed to five seconds, and presented without generation watermarks.

In T1, participants watched and described clips; after a break, T2 required real/AI judgments, with spoken responses transcribed by the experimenter. Each participant viewed 20 clips per phase, half real and half AI, drawn from the full library. Counterbalancing prevented viewing both versions or the same scene across phases. A central fixation display preceded each video. T1 always preceded T2, so phase cannot be separated from order, practice, or fatigue. T1 contributed to eligibility and to the eligibility assessment; the reported behavioral models used T2 only. Gaze was recorded at 60 Hz with a Gazepoint GP3 (no head mount) on a 1680×1050 Samsung SyncMaster monitor at 90 ppi. Participants sat approximately 60–70 cm from the screen. The source study used Gazepoint Control for calibration and Gazepoint Analysis for stimulus presentation and recording. We used the exported fixation records (FPOG fields). The fixed aspect-fit video rectangle was $(140, 0)$ to $(1540, 1050)$ display pixels (1400 display pixels spanning the video width).

3.2 Response reconciliation, registration and eligibility

Fixation, judgment, and presentation-list records were reconciled to explicit video IDs and the 40 study clips, which required unique mappings and agreement among actual class, judged class, and correctness. A trial was one participant's viewing of one clip in one phase. Stimulus tests used 20 scene-pair differences, while behavioral models retained repeated participant and stimulus IDs.

The `Time` field was treated as fixation end time relative to video onset. For endpoint e_k and recorded duration d_k, we intersected $[e_k - d_k, e_k]$ with $[0, 5]$ s. Screen-normalized fixation positions were mapped through the fixed display rectangle to video coordinates. Out-of-video gaze were excluded. Eligible fixations required finite position, time and positive duration, tracker validity, positive within-window overlap, in-video coordinates, and valid interval ordering. Trials required at least five eligible fixations and 2.5 s cumulative in-window duration. Of 787 candidate physics trials, 769 passed (97.7%): 384 T1 and 385 T2. T2 retained 193 actual-AI and 192 real trials from all 40 participants. Extended registration and reconciliation details are in Supplementary Section S1.

3.3 Motion fields and controlled coupling

Feature extraction used presentation time. Each five-second clip was resampled to 24 Hz and 320×240 grayscale pixels, yielding 120 frames and 119 adjacent-frame maps. Maps were pooled by midpoint time onto a common grid with 60 temporal bins and 60×80 spatial bins.

Motion M measured displacement magnitude after estimated global camera movement was removed. Temporal irregularity A combined forward–backward flow disagreement and photometric inconsistency after warping. Neither was validated as a local AI-artifact map. Dense bidirectional optical flow [11] was corrected using a homography fitted by RANSAC [12] to ORB features [25]. Failed registration used uncompensated flow and was logged. Mean *unscaled* motion was retained as a separate covariate. Each clip's motion field was percentile-clipped/scaled, then scaled

motion M was centered and divided by its spatial SD within each time bin. Full parameters and quality checks are in Supplementary Sections S1 and S5.

Bilinear motion samples at each fixation were averaged over overlapping bins with duration weights; the primary score used zero lag. Each fixation was compared with 30 same-time control locations at the same physical distance from the center of the 4:3 frame. In normalized video coordinates,

$$r = \sqrt{[0.8(x - .5)]^2 + [0.6(y - .5)]^2}.$$

Control candidates had to fall inside the video frame and have valid interpolation support across the evaluated lags. Unsupported observed fixations were omitted from the relevant score, with usable count and duration recorded. Controls retained the observed temporal weights and were reused across lags. The generator used seed 20260808 and each trial's `TrialUID`.

For standardized motion $z_t(x, y)$ and overlap seconds w_{kt}, the observed average S and controlled score C were

$$S = \frac{\sum_{k,t} w_{kt} z_t(x_k, y_k)}{\sum_{k,t} w_{kt}}, \qquad C = \frac{S - \overline{S_0}}{\mathrm{sd}(S_{0,1}, \ldots, S_{0,30})}. \tag{1}$$

Each $S_{0,r}$ is a duration-weighted control-replicate score. The sample SD uses denominator $n - 1$; zero SD leaves C undefined. Positive coupling means greater sampled motion than the reference locations. This contrast measures the magnitude of sampled motion, not eye-velocity alignment, verified artifacts, or smooth pursuit.

3.4 Conventional and secondary measures

Scanpath length summed successive fixation-center distances in source-video pixels. Dispersion was the duration-weighted RMS distance from the weighted centroid. Entropy used duration shares p_j on a 10×10 grid, $-\sum_{p_j > 0} p_j \log p_j$. Eligible fixation count and duration described data quantity. These summaries characterize path extent, spread, and occupancy without retaining motion at the moments viewed.

Temporal irregularity A combined forward–backward flow disagreement and photometric inconsistency after warping, with per-clip percentile scaling and a validity mask. The exploratory irregularity-coupling score applied Eq. 1 to A. Video H_0 used cubical sublevel persistence on $1 - A$, with unsupported cells and a one-voxel border entering at filtration value 1, coefficients in $\mathbb{Z}_2$, and reduced homology [4, 28]. Total persistence summed finite positive lifetimes $P_{H_0}^{\mathrm{video}} = \sum_i (d_i - b_i)$. Lifetimes are threshold differences, not event durations. This describes normalized processed-field structure, not physical plausibility. Exploratory gaze H_0 applied the same construction to separately normalized, smoothed, duration-weighted fixation fields. Its addition to judgment models used $\log(1 + P_{H_0}^{\mathrm{gaze}})$. Supplementary Section S5 gives construction and diagnostics.

Brief experimenter-transcribed reasons were available on 160 of 161 eligible S1 AI responses: 122 hits and 38 false alarms. Two language-model coders independently assigned overlapping cue labels to anonymized text-only reports after practice feedback; a human reviewer adjudicated reports with a disagreement. Labels describe stated information, not verified video cues. A concrete-cue report had at least one dynamics, physical-event, object-change, or appearance label; impression-only was its complement. Dynamics indicated movement/temporal behavior and physical-event labels indicated physical/causal implausibility. Inter-model agreement does not establish coding accuracy against independent ground truth. Because this sample conditions on AI responses, hits versus false alarms also contrasts actual-AI versus real clips. Full coding, agreement, and model details are in Supplementary Section S7.

3.5 Statistical analysis

The primary logistic mixed model used maximum-likelihood `glmer` in `lme4` [2], Laplace approximation and `bobyqa`. AI judgment was 1, Real judgment 0, with actual-real as reference. Coupling, motion, fixation count, and duration were standardized across the same 385 T2 trials. The fixed-effects expression was

$$\operatorname{logit}\Pr(J = \mathrm{AI}) = \beta_0 + \beta_1 z(C) + \beta_2 a + \beta_3 z(C)a + \gamma_1 z(\bar{m}) + \gamma_2 z(N) + \gamma_3 z(D). \tag{2}$$

Participant and video had random intercepts. Here C is the controlled coupling score, $a = 1$ denotes actual AI, $\bar{m}$ mean unscaled motion, N fixation count, and D duration. Conditional coupling slopes are β_1 (real) and $\beta_1 + \beta_3$ (AI), with covariance-based standard errors; $\exp(\beta_3)$ is their ratio of odds ratios (ORs). Tests and 95% intervals were two-sided Wald estimates.

The nested comparisons used the same 385 rows, T2-wide standardization, and participant/video intercepts as the primary model. All models included actual class, mean motion, fixation count, and duration. *Adjusted coupling* was the primary model in Eq. 2. *Conventional* added scanpath length, dispersion, and entropy to the common adjustments; *Combined* contained these descriptors together with coupling and its class interaction. A sensitivity allowed all five conventional gaze measures to have class-specific slopes before adding coupling and its interaction. Likelihood-ratio (LR) tests assessed nested additions; AIC/BIC compared specified models. The 20 scene-level AI-minus-real differences were units for paired clip tests, with 95% t intervals and 19 df. Reason profiles used participant intercepts; focal reason–gaze models adjusted for actual class, with fit limitations reported alongside results. Tests across this selected secondary analysis were nominal and unadjusted. Supplementary Sections S3–S7 give extended models and diagnostics.

3.6 Reference and timing sensitivities

We compared the primary spatial reference with uniform controls and motion sampled 100 or 200 ms earlier. An exploratory transition reference adapted the saccadic-flow approach [6] using a joint truncated Gaussian, mirrored training data, and robust fourth-order surfaces for its two location, two scale, and correlation parameters. Task-matched fits pooled actual classes, balanced participant–scene cells, and excluded the evaluated scene pair. Wider local windows were necessary for this cohort. The mirrored quartic was selected among three candidates by internal held-out endpoint log density before judgment fitting. It is an adaptation rather than a reproduction. Joint-reference and rescored eccentricity coupling used the same 7,386 transitions across T1/T2 and 30 controls per transition, retaining all 385 T2 trials. The joint-reference judgment model used participant/scene intercepts; its random-structure diagnostics are reported in Supplementary Section S2.

A separate exploratory, content-aware comparison applied frozen DeepGaze III [16] to a launch-time RGB frame and up to four preceding fixation locations on a common history-eligible subset. It retained all 385 T2 trials and used 300 controls per transition for each reference, with participant/scene intercepts. Supplementary Section S2 gives the implementation, matched comparisons, and calibration checks. Each coupling was standardized separately within T2; separate reference fits do not test coefficient differences, and reference-fitting uncertainty was not propagated.

3.7 Software and reproducibility

Analyses used Python 3.12.7 and R 4.4.2, with `lme4`, `lmerTest` [17], and giotto-tda. Configuration, package versions, input hashes, run commands, and supporting analyses are described in Supplementary Section S8.

4 Results

4.1 Motion coupling and authenticity judgments

Among the 385 T2 trials, 38 of 192 actual-real trials were judged AI (false alarms) and 123 of 193 actual-AI trials were judged AI (correct detections). Mean motion coupling across all T2 trials was near zero (+0.12 control SDs, SE = 0.25, $p = .640$), so viewers did not, on average, sample more motion than eccentricity-matched controls. Raw means were +0.48 on real clips and −0.25 on AI clips; the class contrast in standardized units was −0.40 SD (AI minus real; $p = .045$ in a singular three-intercept fit).

In the primary adjusted all-T2 model with participant and video random intercepts, the coupling slope differed by actual authenticity: $\beta = 1.139$ (SE = 0.360), ratio of odds ratios = 3.12 (95% CI [1.54, 6.33]), $p = .00158$ (Figure 2). The actual-real slope was $\beta = -0.550$, AI-response OR = 0.58 [0.36, 0.94]), $p = .0265$; the actual-AI slope was $\beta = 0.589$, OR = 1.80 ([1.08, 3.00]), $p = .0240$. Thus, relative to actual clip class, higher coupling tracked more accurate judgments. These estimates use one coupling SD across T2 and describe associations conditional on actual class.

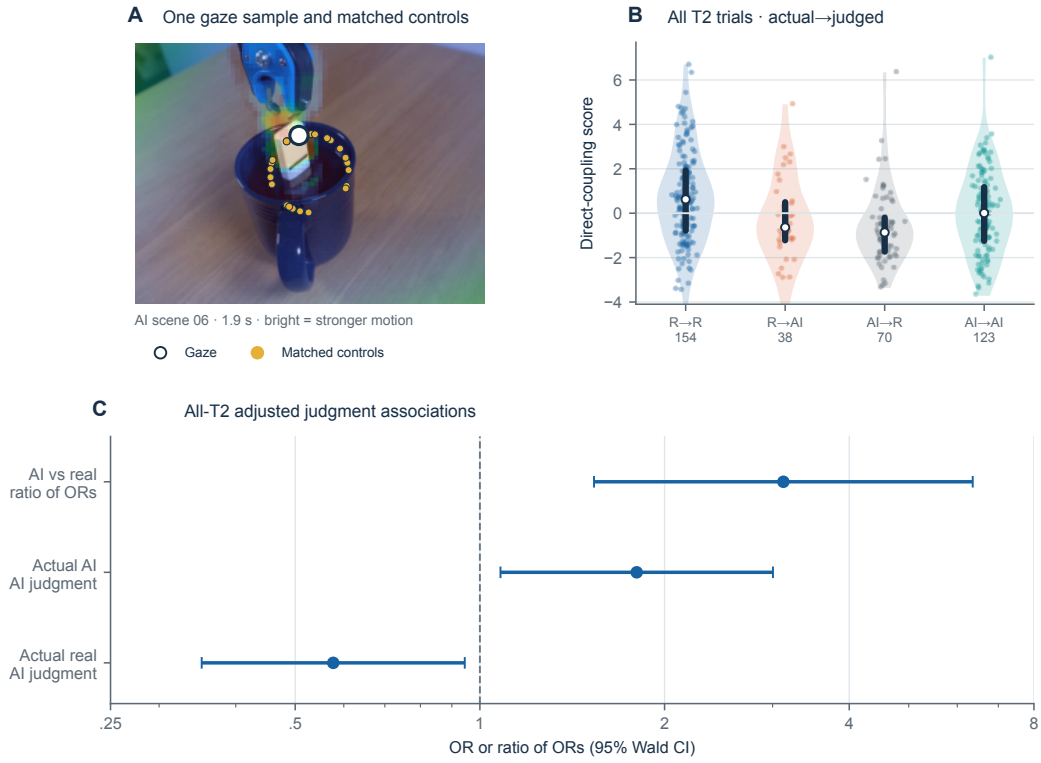


Fig. 2. **Direct gaze–motion coupling and AI judgments. A**, Example AI frame at 1.9 s: brighter motion overlay denotes stronger motion; white marks a fixation and gold its 30 same-time, eccentricity-matched controls. **B**, Scores for all 385 T2 trials, grouped by actual → judged authenticity. **C**, Adjusted AI-response odds ratios within actual-real and actual-AI clips, and their ratio, with 95% Wald intervals from the primary three-intercept model.

Table 1. Exploratory reference sensitivity on 7,386 common transition contributions across T1/T2; each judgment fit uses 385 T2 trials, 30 controls, and lag 0. AI-response ORs are per within-T2 SD of the reference-specific coupling score. Brackets show 95% Wald intervals.

Reference	Actual real OR	Actual AI OR	Ratio of ORs
Eccentricity (rescored)	.54 [.32,.90]	1.66 [1.02,2.72]	3.07 [1.51,6.23]
Joint saccadic-flow adaptation	.46 [.28,.76]	1.20 [.80,1.79]	2.60 [1.38,4.91]

Eccentricity uses participant/scene/video intercepts; the joint reference uses participant/scene intercepts after a singular three-intercept fit. Scores exclude first and otherwise ineligible transitions. Intervals condition on estimated reference and control scores; separate fits do not test differences between reference coefficients.

Sensitivity to reference and timing. The class interaction remained under uniform controls (ratio of ORs = 3.14, 95% CI $[1.48, 6.65]$, $p = .00286$) and 100- and 200-ms earlier motion fields (ratios 3.46 and 3.37; both $p < .001$). On the common transition subset with 30 controls, the rescored eccentricity interaction was 3.07 ($[1.51, 6.23]$), $p = .0019$. Under the mirrored joint saccadic-flow adaptation it was 2.60 ($[1.38, 4.91]$), $p = .0032$ (Table 1). The real-clip slope was negative (OR = .46, $[.28, .76]$), whereas the AI-clip slope was imprecise (OR = 1.20, $[.80, 1.79]$). These estimates retain evidence of a class interaction under the joint reference without establishing a positive AI-clip slope.

In the separate history-eligible comparison, the frozen-frame DeepGaze III interaction was 1.69 ($[.92, 3.13]$), $p = .093$. Its interval includes no interaction and substantial positive interactions. Separate fits do not establish a difference from the primary estimate. Full implementation, matched-reference comparisons, and imperfect endpoint calibration are reported in Supplementary Section S2.

Comparison with conventional summaries. The participant/video models had AICs of 421.97 (Adjusted coupling), 434.32 (Conventional), and 427.90 (Combined). Adding motion coupling and its class interaction to Conventional improved fitted likelihood, LR $\chi^2(2) = 10.42$, $p = .00546$, $\Delta\text{AIC} = -6.42$. This improvement remained when all five conventional gaze measures had class-specific slopes, $\chi^2(2) = 10.39$, $p = .00555$, $\Delta\text{AIC} = -6.39$. Adding scanpath length, dispersion, and entropy to Adjusted coupling supplied no clear improvement, $\chi^2(3) = 0.07$, $p = .995$, $\Delta\text{AIC} = +5.93$. Adjusted coupling had the lowest AIC/BIC among these specified models. Complete formulas, estimates, and random-structure checks are in Supplementary Section S3.

4.2 Exploratory comparisons

Irregularity coupling showed a weaker class interaction under eccentricity controls (ratio of ORs = 2.08, $[1.17, 3.71]$, $p = .0126$), which was absent under uniform controls (0.98, $[0.56, 1.72]$, $p = .952$). In the joint participant/scene fit, the motion interaction remained (2.91, $[1.49, 5.68]$, $p = .00178$), whereas irregularity was imprecise (1.44, $[.78, 2.67]$, $p = .243$). In separate participant/scene models containing the conventional measures and motion coupling, adding irregularity gave LR $\chi^2(2) = 1.51$, $p = .471$; adding gaze H_0 and its class interaction gave $\chi^2(2) = 1.61$, $p = .446$. These supportive comparisons supplied no clear incremental judgment information from those measures (Supplementary Sections S3 and S6).

Across 20 matched scenes, mean video H_0 persistence was 206.48 for AI and 374.78 for real clips. The paired difference was −168.30 (95% CI $[-302.81, -33.80]$), $t(19) = -2.62$, $p = .0169$. Scenes 04/07/09 supplied 56.7% of absolute differences; omitting them yielded a smaller difference of −69.76 ($[-127.48, -12.04]$). Unresolved flow/registration checks and per-clip scaling limit interpretation. This is a contrast in normalized processed fields, not a physical-realism judgment. Detailed clip-property, between-viewer, and topology results appear in Supplementary Sections S5–S6.

Concrete cues occurred in 98/122 hit reports (80.3%) and 22/38 false alarms (57.9%): participant-intercept OR 4.98 ([1.72, 14.42]), $p = .003$. Because correctness and actual class coincide within AI responses, this cannot isolate successful detection from stimulus properties. Dynamics reports did not accompany clearly higher motion coupling after class adjustment ($b = -.12$ SD, $[-.45, .20]$, $p = .455$); the dynamics/physical-event composite was similarly inconclusive ($b = -.10$, $[-.42, .21]$, $p = .512$). Both fits retained a singular participant-only intercept and no video intercept. These estimates do not establish equivalence or a dissociation between reported cues and gaze sampling. Supplementary Section S7 reports the full coding and exploratory analyses.

5 Discussion

5.1 Motion sampling and the interpretation of gaze

Across 385 T2 trials, higher gaze–motion coupling was associated with fewer AI judgments for real clips and more AI judgments for AI clips under the primary eccentricity-matched reference. Both directions correspond to more accurate judgments relative to actual clip class. This result connects gaze sampling to the participant's response while retaining the motion available at the particular locations and moments viewed, extending the source study's broad gaze comparisons [26]. The finding does not establish a general gaze signature of generated content, consistent with the diversity of tasks and findings in prior authenticity research [24].

Although mean coupling was close to zero overall, our main question concerned how variation in coupling related to authenticity judgments. The mean describes average excess motion sampling relative to the controls, whereas the coupling–judgment slopes describe how differences across trials relate to responses within each actual clip class. Viewers therefore need not show an overall preference for higher-motion locations for relative motion sampling to relate to their judgments. Separately, adding coupling and its actual-class interaction improved judgment-model fit beyond conventional gaze summaries, including when those summaries had class-specific slopes. This supports the value of retaining the motion available at fixation locations and times in this sample; it does not establish that conventional summaries are generally uninformative.

Under the primary reference, higher coupling accompanied more accurate judgments within both actual clip classes. One possible explanation is that viewers use motion encountered at fixation to assess whether the depicted event unfolds plausibly. Motion consistent with physical expectations could support a Real response, whereas visible inconsistencies could support an AI response. Prior work shows that motion and task can influence gaze allocation [19, 27], and viewers report motion among authenticity cues [7]. However, our analysis does not identify whether the sampled motion was realistic, anomalous, or diagnostically useful. Motion was not annotated for diagnosticity, and camera, object, and rendering changes may contribute differently. Alternatively, viewers already leaning toward a judgment may sample different regions. Without time-resolved decisions, independent artifact labels, or an intervention on viewing, these explanations remain unresolved.

5.2 What the reference sensitivity establishes

The control reference defines the sampling comparison. Eccentricity matching asks whether fixations sample more motion than alternative positions at the same distance from the center. The joint transition reference instead uses the preceding fixation to define plausible endpoints. The positive class interaction under these references reduces dependence on one location control. Though the joint reference retained the class interaction, its imprecise AI-clip

slope means that this analysis does not clearly establish an association between higher coupling and more accurate judgments for AI clip class.

The joint reference is fitted to other scenes from the same task and cohort. It therefore provides a task-matched empirical comparison rather than a pure measure of image-independent oculomotor behavior. Scene exclusion prevents the evaluated pair from training its own reference, but shared task and scene-composition tendencies may remain. The joint model improved internal held-out density over the earlier axiswise reference, yet its wider windows, model selection on the same evaluation set, and imperfect calibration limit that evidence. Predictive improvement does not show that all relevant movement biases have been captured.

Evidence for the interaction was less clear under the content-aware DeepGaze reference. One possible explanation concerns how the reference defines expected motion sampling. DeepGaze uses a static video frame and preceding fixation locations to predict where a participant might look next. Although it receives no explicit motion input, the visually prominent regions it favours may also contain motion. When control locations are sampled from this distribution, the actual video motion at those locations contributes to the expected baseline. This could leave less excess motion at observed gaze locations—that is, a smaller difference between the motion participants sampled and the motion expected under the reference. However, we have not established whether this mechanism explains the smaller class-interaction estimate, which measures how the association between motion coupling and AI judgments differs between real and AI-generated clips. A reduction in excess motion does not necessarily reduce that interaction. Furthermore, the references differ in model structure, training and history representation. DeepGaze's predicted fixation probabilities were also imperfectly calibrated: regions assigned 90% probability contained approximately 74% of observed fixation endpoints. This indicates a mismatch between predicted and observed gaze distributions. Together, these limitations prevent attributing the smaller interaction estimate specifically to content conditioning.

5.3 Secondary measures and stated cues

Irregularity coupling depended more strongly on the spatial reference and added little after motion was included. Video-field persistence also distinguished the matched clip classes, but the field combines flow and photometric consistency after separate per-clip normalization. A difference in its connected-component lifetimes need not indicate greater physical plausibility, more artifacts, or information used by a viewer. The absence of a clear improvement in judgment-model fit from adding gaze-H_0 likewise limits claims about its additional value for modeling authenticity judgments in this sample, without establishing that topological representations lack value in other tasks.

Concrete cues were more common in reports accompanying AI hits than in those accompanying false alarms, yet these groups also differ in the actual stimulus class. The reason–gaze estimates did not clearly link dynamics reports to greater motion sampling. Brief retrospective reports and online gaze measurements operate at different levels: a participant can mention an event without fixating on its most motion-rich location, and motion can be sampled without being named afterward. The present intervals and clustering limitations leave several such accounts compatible with the data. Collecting explanations after both Real and AI judgments, including correct and incorrect responses, and linking these reports to independently annotated video events would make this comparison more informative.

5.4 Limitations and implications for follow-up studies

The 385 T2 trials come from the same 40 participants and 20 matched physics-scene pairs. They therefore provide repeated observations within a limited participant and stimulus sample, rather than independent replications. The real and AI versions depict corresponding events but differ in their visual realization, so class-related differences cannot be

attributed solely to generation. The physics domain and single generator also restrict generalization to other content and generation systems. T1 always preceded T2, preventing task differences from being separated from order, practice, or fatigue. Moreover, actual clip class enters the judgment models: the fitted associations and held-out diagnostics concern viewers' responses conditional on known class, rather than detecting the authenticity of an unknown clip.

Tests across the selected comparisons were unadjusted. The primary model and its nested comparisons used participant/video intercepts, while some supporting analyses used participant/scene intercepts. Separate fits do not formally test differences between reference-specific effects. Reported intervals also condition on the estimated transition references and sampled control locations; they do not include uncertainty from fitting those references or repeating the control sampling. The sensitivity results therefore assess dependence on particular analysis choices within this cohort, rather than provide independent confirmation.

Because the analysis uses exported fixation summaries, coupling describes motion sampled at recorded gaze locations; it does not establish smooth pursuit or eye–motion synchrony. Registration failures left some frame pairs with uncompensated flow, while extreme motion estimates and near-static clips warrant further inspection of the flow fields and their normalization. All reference analyses reuse these motion fields, so consistency across references does not independently validate the motion estimates or exclude errors shared across analyses.

A follow-up should prespecify the primary coupling measure and class interaction, inspect optical-flow estimates against the source videos, and recruit new participants, scenes, and generators to assess generalization. Linking gaze to independently annotated motion events and visible inconsistencies would help identify which content viewers sampled. Experimentally manipulating access to regions independently identified as diagnostic would then provide a stronger test of whether information from those regions contributes to authenticity judgments. Similarly, recording when participants report committing to a Real or AI judgment would also allow gaze sampling before and after reported commitment to be compared.

6 Conclusion

We found that how viewers sampled motion related to their authenticity judgments in this physics-video task. Under the primary eccentricity-matched reference, higher gaze–motion coupling accompanied more accurate judgments for both real and AI clips. Coupling also improved judgment-model fit beyond conventional gaze summaries, including when their slopes could differ by actual clip class. The class interaction persisted under the joint transition reference, although the AI-clip slope was uncertain. The DeepGaze III comparison gave less clear evidence for the interaction. Our analysis connects viewers' judgments to the relative motion they sampled at particular locations and moments. We cannot tell whether motion sampling helped shape viewers' judgments, their developing judgments guided motion sampling, or both.

Privacy and Ethics Statement

The original study had ethics approval and informed consent, as reported by Shi et al. [26]; this secondary analysis used de-identified fixation and response records without new recruitment.

Data Availability Statement

An anonymized link for the de-identified derived data and analysis code is available here at Figshare.

References

[1] Ioannis Agtzidis, Mikhail Startsev, and Michael Dorr. 2016. Smooth pursuit detection based on multiple observers. In *Proceedings of the symposium on eye tracking research and applications*. 303–306.

[2] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. 2015. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software* 67, 1 (2015), 1–48. doi:10.18637/jss.v067.i01

[3] Babette Bühler, Efe Bozkir, Hannah Deininger, Peter Gerjets, Ulrich Trautwein, and Enkelejda Kasneci. 2024. On Task and in Sync: Examining the Relationship between Gaze Synchrony and Self-Reported Attention During Video Lecture Learning. *Proceedings of the ACM on Human-Computer Interaction* 8, ETRA, Article 230 (2024), 18 pages. doi:10.1145/3655604

[4] Gunnar Carlsson. 2009. Topology and data. *Bull. Amer. Math. Soc.* 46, 2 (2009), 255–308. doi:10.1090/S0273-0979-09-01249-X

[5] Ran Carmi and Laurent Itti. 2006. Visual causes versus correlates of attentional selection in dynamic scenes. *Vision Research* 46, 26 (2006), 4333–4345. doi:10.1016/j.visres.2006.08.019

[6] Alasdair D. F. Clarke, Matthew J. Stainer, Benjamin W. Tatler, and Amelia R. Hunt. 2017. The saccadic flow baseline: Accounting for image-independent biases in fixation behavior. *Journal of Vision* 17, 11 (2017), 12, 1–19. doi:10.1167/17.11.12

[7] Catherine E. Davodi, Sarah Barrington, Hany Farid, and Emily A. Cooper. 2026. Perceptual Judgments of Video Authenticity: An Examination of Viewing Duration, Confidence, Content, and Strategies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. 10679–10686. https://openaccess.thecvf.com/content/CVPR2026W/APAI/html/Davodi_Perceptual_Judgments_of_Video_Authenticity_An_Examination_of_Viewing_Duration_CVPRW_2026_paper.html

[8] Ilke Demir and Umur Aybars Ciftci. 2021. Where do deep fakes look? synthetic face detection via gaze tracking. In *Proceedings of the symposium on eye tracking research and applications*. 1–11.

[9] Michael Dorr, Thomas Martinetz, Karl R. Gegenfurtner, and Erhardt Barth. 2010. Variability of eye movements when viewing dynamic natural scenes. *Journal of Vision* 10, 10 (2010), 28, 1–17. doi:10.1167/10.10.28

[10] K. Anders Ericsson and Herbert A. Simon. 1993. *Protocol Analysis: Verbal Reports as Data* (revised ed.). The MIT Press, Cambridge, MA. https://mitpress.mit.edu/9780262050470/protocol-analysis/

[11] Gunnar Farnebäck. 2003. Two-Frame Motion Estimation Based on Polynomial Expansion. In *Image Analysis (Lecture Notes in Computer Science, Vol. 2749)*. Springer, 363–370. doi:10.1007/3-540-45103-X_50

[12] Martin A. Fischler and Robert C. Bolles. 1981. Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography. *Commun. ACM* 24, 6 (1981), 381–395. doi:10.1145/358669.358692

[13] Oliver Hein and Wolfgang H. Zangemeister. 2017. Topology for gaze analyses - Raw data segmentation. *Journal of Eye Movement Research* 10, 1 (2017), 1, 1–25. doi:10.16910/jemr.10.1.1

[14] Marius Huber, David Robert Reich, and Lena Ann Jäger. 2026. Fixation Sequences as Time Series: A Topological Approach to Dyslexia Detection. In *Proceedings of the 2026 Symposium on Eye Tracking Research and Applications*. 1–13.

[15] Petter Johansson, Lars Hall, Sverker Sikström, and Andreas Olsson. 2005. Failure to Detect Mismatches Between Intention and Outcome in a Simple Decision Task. *Science* 310, 5745 (2005), 116–119. doi:10.1126/science.1111709

[16] Matthias Kümmerer, Matthias Bethge, and Thomas S. A. Wallis. 2022. DeepGaze III: Modeling free-viewing human scanpaths with deep learning. *Journal of Vision* 22, 5 (2022), 7, 1–27. doi:10.1167/jov.22.5.7

[17] Alexandra Kuznetsova, Per B. Brockhoff, and Rune H. B. Christensen. 2017. lmerTest Package: Tests in Linear Mixed Effects Models. *Journal of Statistical Software* 82, 13 (2017), 1–26. doi:10.18637/jss.v082.i13

[18] Dominique Makowski, An Shu Te, Ana Neves, Stephanie Kirk, Ngoi Zi Liang, Panagiotis Mavros, and S. H. Annabel Chen. 2025. Too beautiful to be fake: Attractive faces are less likely to be judged as artificially generated. *Acta Psychologica* 252 (2025), 104670. doi:10.1016/j.actpsy.2024.104670

[19] Parag K. Mital, Tim J. Smith, Robin L. Hill, and John M. Henderson. 2011. Clustering of gaze during dynamic scene viewing is predicted by motion. *Cognitive Computation* 3, 1 (2011), 5–24. doi:10.1007/s12559-010-9074-z

[20] Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. 2026. Do Generative Video Models Understand Physical Principles?. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 948–958. doi:10.1109/WACV61042.2026.00099

[21] Richard E. Nisbett and Timothy D. Wilson. 1977. Telling More Than We Can Know: Verbal Reports on Mental Processes. *Psychological Review* 84, 3 (1977), 231–259. doi:10.1037/0033-295X.84.3.231

[22] Nikolas Papadopoulos, Shreenithi Navaneethan, Sheng Bai, Ankur Samanta, and Paul Sajda. 2026. Gaze patterns predict preference and confidence in pairwise AI image evaluation. In *Proceedings of the 2026 ACM Symposium on Eye Tracking Research & Applications*. Association for Computing Machinery, New York, NY, USA, 15:1–15:7. doi:10.1145/3797246.3803031

[23] Petra M. J. Pollux, Matthew Craddock, and Kun Guo. 2019. Gaze patterns in viewing static and dynamic body expressions. *Acta Psychologica* 198 (2019), 102862. doi:10.1016/j.actpsy.2019.05.014

[24] Hakile Resulbegoviq, Valentina Hlebec, Mirjana Pejić Bach, and Irina Stamatović. 2026. Eye Tracking and AI-Generated Content: A Systematic Literature Review of Visual Attention, Cognitive Processing, and User Engagement. *Journal of Eye Movement Research* 19, 5 (2026), 97. doi:10.3390/jemr19050097

[25] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. 2011. ORB: An Efficient Alternative to SIFT or SURF. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2564–2571. doi:10.1109/ICCV.2011.6126544

[26] Danqing Shi, Lan Jiang, Katherine M. Collins, Shangzhe Wu, Ayush Tewari, and Miri Zilka. 2026. Perceived authenticity drives gaze behavior when watching AI-generated videos of physical scenes. *Scientific Reports* 16 (2026), 25792. doi:10.1038/s41598-026-55437-5

[27] Tim J. Smith and Parag K. Mital. 2013. Attentional synchrony and the influence of viewing task on gaze behavior in static and dynamic scenes. *Journal of Vision* 13, 8 (2013), 16. doi:10.1167/13.8.16

[28] Guillaume Tauzin, Umberto Lupo, Lewis Tunstall, Julian Burella Pérez, Matteo Caorsi, Anibal M. Medina-Mardones, Alberto Dassatti, and Kathryn Hess. 2021. giotto-tda: A topological data analysis toolkit for machine learning and data exploration. *Journal of Machine Learning Research* 22, 39 (2021), 1–6.

[29] Geoffrey Tien, M Stella Atkins, and Bin Zheng. 2012. Measuring gaze overlap on videos between multiple observers. In *Proceedings of the symposium on eye tracking research and applications*. 309–312.

[30] Po-He Tseng, Ran Carmi, Ian G. M. Cameron, Douglas P. Munoz, and Laurent Itti. 2009. Quantifying center bias of observers in free viewing of dynamic natural scenes. *Journal of Vision* 9, 7 (2009), 4, 1–16. doi:10.1167/9.7.4

[31] David C Wong, Bin Wang, Gorkem Durak, Marouane Tliba, Mohamed Amine Kerkouri, Aladine Chetouani, Ahmet Enis Cetin, Cagdas Topel, Nicolo Gennaro, Camila Vendrami, et al. 2025. Shifts in Doctors' Eye Movements Between Real and AI-Generated Medical Images. In *Proceedings of the 2025 Symposium on Eye Tracking Research and Applications*. 1–7.

Supplementary Material: Gaze Sampling of Motion Relates Differently to Authenticity Judgments for Real and AI-Generated Physics Videos

ANONYMOUS AUTHOR(S)

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Computer vision.*

Additional Key Words and Phrases: eye tracking, dynamic scene viewing, generative video, visual attention, gaze–motion coupling, persistent homology, supplementary methods

This document provides supporting implementation details, diagnostics, and exploratory analyses for the main paper. The main contribution and its principal qualifications are reported in the main paper. Sections S1–S8 are supplementary sections. The analysis uses the physics-video subset of Shi et al.'s eye-tracking study [Shi et al. 2026].

S1 Extended preprocessing and location controls

S1.1 Response reconciliation and analytic units

Fixation, judgment, and presentation-list records were reconciled to explicit video IDs and the 40 study clips. The final ID suffix encoded actual class (-a for AI, -r for real); removing it gave the shared scene ID for each matched pair. The reconciliation required unique presentation/video and trial/response mappings, agreement on video assignment, and correctness equal to agreement of actual and judged class.

The 385 eligible T2 trials analyzed below use these labels. A trial was one participant's viewing of one clip in one phase; stimulus tests used 20 scene-pair differences, while behavioral models retained repeated participant and stimulus IDs.

S1.2 Registering fixations to the five-second video

The `Time` field was treated as fixation end time relative to video onset. For endpoint e_k and recorded duration d_k (`FPOGD`), we intersected $[e_k - d_k, e_k]$ with $[0, 5]$ s. Fixation indices were required to be consecutive from 1; a violation would halt preprocessing. Among finite, positive-duration intervals, trials were designated for quarantine if fixation end times were non-increasing or an interval began more than 0.0001 s before the preceding interval ended. No trials met these interval-quarantine criteria in the analyzed run. Fixations longer than 2 s were flagged but not automatically excluded. Screen-normalized `FPOGX/FPOGY` were converted through display pixels to video coordinates.

We required $0 \le x, y < 1$, excluding out-of-video gaze. Distance summaries used 960×720 source-video pixels. Eligible fixations had finite position/end time/duration, positive duration, tracker validity `FPOGV=1`, positive within-window overlap, in-video coordinates, and valid interval ordering. Trials required at least five eligible fixations and 2.5 s cumulative in-window duration. Of 787 candidate physics trials, 769 passed these criteria (97.7%): 384 T1 and 385 T2. The remaining 18 trials (2.3%) failed eligibility. T2 retained 193 actual-AI and 192 real trials from all 40 participants.

S1.3 Video standardization and common feature grid

Feature extraction used presentation time rather than source frame index. The manifest checked the declared 40-video library, scene pairing, monotonically increasing decoded timestamps, source hashes, and the five-second window.

FFmpeg cloned the last frame for at most 0.06 s when an acceptable encode ended slightly early, trimmed to five seconds, resampled to 24 Hz, resized with Lanczos interpolation to 320×240, converted to grayscale, and wrote lossless FFV1 derivatives. Each derivative had exactly 120 frames. Video and gaze fields shared 60 temporal bins and 60×80 spatial bins: 0.0833 s and a 4×4-pixel derivative block per cell. Resampling each five-second clip to 24 Hz yielded 120 frames and 119 adjacent-frame maps. Each map was assigned to one of 60 temporal bins using the midpoint time of its frame pair; within each time bin, finite measurements were averaged over each 4×4-pixel spatial cell, omitting invalid samples.

This fixed grid prevented source encoding rate from determining the number of measurements in the topology summary.

S1.4 Motion and temporal-irregularity fields

Motion M measured displacement magnitude after estimated global camera movement was removed. Temporal irregularity A combined forward–backward flow disagreement and photometric inconsistency after warping. Neither was validated as a local AI-artifact map by human raters. Dense bidirectional optical flow followed Farnebäck [Farnebäck 2003] (pyramid scale .5, three levels, a 15-pixel window, three iterations, polynomial neighborhood five, and smoothing 1.2). Camera displacement came from a homography fitted by RANSAC [Fischler and Bolles 1981] (three-pixel threshold), using up to 1000 ORB features [Rublee et al. 2011] and the 300 best cross-checked matches, requiring eight matches and six inliers. Failed registration used uncompensated flow and was logged. Local-motion magnitude was the norm of forward flow minus homography displacement, multiplied by 24 (derivative pixels/s).

Consistency used the *total* forward-flow map: forward–backward error was $\|\mathbf{f}(\mathbf{x}) + \mathbf{b}(\mathbf{x} + \mathbf{f}(\mathbf{x}))\|$, and photometric error was the absolute gray-level difference after that warp. Out-of-frame warps were invalid; pooled irregularity required 50% finite transport support. Per video, motion and each irregularity component were clipped/scaled using the first and 99th percentiles over supported voxels. The equal-weight irregularity mean was scaled again to form $A \in [0, 1]$. Unsupported A voxels were zero-filled with a separate validity mask. Mean *unscaled* motion was retained as a covariate, separate from controlled coupling.

S1.5 Direct gaze–motion coupling and location controls

Within each time bin, scaled motion M was centered and divided by its spatial SD over valid cells. Bilinear samples at each fixation were averaged over overlapping bins with duration weights; the primary score used zero lag. Each fixation was compared with 30 same-time control locations at the same physical distance from the center of the 4:3 video frame. Using normalized video coordinates (x, y), we calculated this distance as

$$r = \sqrt{[0.8(x - .5)]^2 + [0.6(y - .5)]^2}.$$

For each fixation, we sampled 30 control locations at the same physical radius r. A control candidate had to fall inside the video frame and have valid feature support: every grid cell used by bilinear interpolation had to be supported at all time bins needed for that fixation across the evaluated lags. If a candidate failed either check, the sampler tried another; the fixation itself was not discarded for that reason. An observed fixation was omitted from a lag-specific score only if its own interpolation cells lacked support at a contributing time bin. We recorded the usable fixation count and cumulative overlap duration for each trial and lag.

The pseudorandom generator used seed 20260808 and each trial's `TrialUID`. In each of the 30 control replicates, each fixation's location was replaced by its corresponding control location, while retaining the same temporal overlap weights w_{kt} as the observed score. Control locations were reused across the lag analyses.

For standardized motion $z_t(x, y)$ and overlap seconds w_{kt}, the observed average S and controlled score C were

$$S = \frac{\sum_{k,t} w_{kt} z_t(x_k, y_k)}{\sum_{k,t} w_{kt}}, \qquad C = \frac{S - \overline{S_0}}{\mathrm{sd}(S_{0,1}, \ldots, S_{0,30})}. \tag{S1}$$

Each $S_{0,r}$ is the score for control replicate r, using the same fixation-specific temporal weights as S. The standard deviation is the sample SD across the 30 control scores (denominator $n - 1$); if it is zero, C is undefined. This same-video, same-time, same-eccentricity contrast measures sampled motion magnitude.

The exploratory irregularity score applies the same formula to A, using its feature-specific support mask and saved controls.

S2 Gaze-transition and content-aware reference sensitivities

S2.1 Analysis samples and comparability

Two exploratory analyses assessed alternative gaze references. Both retained the same 769 trials (384 T1 and 385 T2), but used different subsets of fixations within those trials. A trial denotes a participant's viewing of a single clip in a single task. The first analysis evaluated the joint saccadic-flow reference. A total of 7,598 transitions between consecutive valid fixations were eligible for reference fitting. Of these, 7,386 contributed to the trial-level motion-coupling scores after applying the additional motion-scoring criteria. This joint reference replaces the simplified axiswise reference as the principal transition-based sensitivity analysis reported in the main paper. The second analysis compared Gaussian and DeepGaze III references on a common sample with sufficient fixation history. Each evaluated endpoint required at least two consecutive preceding valid fixations, producing a smaller sample than the first analysis. Gaze-prediction performance was evaluated on 6,756 endpoints (3,341 T1; 3,415 T2). Of these, 6,601 also met the existing motion-scoring criteria and contributed to the trial-level coupling scores (3,254 T1; 3,347 T2). All references within this benchmark used the same endpoint-evaluation sample and the same motion-scoring sample. Thus, retaining the same trials did not mean retaining the same fixation contributions: a trial could remain in both analyses while its coupling score was calculated from different subsets of fixations. Comparisons between references within the second benchmark used matched samples. Although both analyses included the same 769 trials, they used different fixation samples, so comparisons across analyses may reflect both sample composition and reference choice. All judgment models used the 385 T2 trials and included actual clip class (real or AI-generated) as a predictor.

S2.2 Joint saccadic-flow adaptation

Clarke et al. [Clarke et al. 2017] modeled next-fixation endpoints conditional on launch position using local joint truncated-Gaussian fits and robust fourth-order regressions. Our adaptation used task-matched, participant–scene-balanced fits pooled across actual classes and trained excluding both versions of the evaluated scene pair. Judgment labels and motion-field values did not enter reference fitting. Other scenes from an evaluated participant could contribute to training; this is scene exclusion rather than participant exclusion.

Image coordinates were mapped to $[-1, 1] \times [-.75, .75]$. For launch position s and endpoint e, the reference density was

$$p(e \mid s) = \frac{\phi_2(e; \mu(s), \Sigma(s))}{\Pr\{\mathcal{N}_2(\mu(s), \Sigma(s)) \in [-1, 1] \times [-.75, .75]\}}. \tag{S2}$$

At 70 launch-position centers on a 10×7 lattice, weighted truncated likelihood estimated two Gaussian locations, two scales, and a correlation. Windows began at width .2 and expanded along a fixed schedule until they contained at least 250 rows, 125 distinct source transitions, Kish effective distinct-source size 100 [Kish 1965], eight scenes, and 12

participants. Reflected weights were aggregated by original transition for the Kish calculation; reflected copies were not treated as independent observations. The effective size diagnoses weight concentration rather than correcting repeated-measure dependence. Windows overlap.

Widening was necessary: no width-.05 window on the published .01 grid met the 250-row threshold in any of the 40 task-by-scene training folds, even after mirroring. Median widths were .8 without mirroring and .4 with mirroring in these frame units, with maxima 1.5 and 1.0. These are substantially smoothed local fits rather than fine spatial measurements. Huber robust surfaces of total degree two or four were fitted to the two locations, log scales, and Fisher-transformed correlation, with ridge 10^{-4}. These transforms preserved positive-definite covariance. Mirroring added horizontal, vertical, and jointly reflected transitions, each of the four copies receiving one-quarter weight. No local fit hit its parameter bounds, and no held-out prediction required clipping. Quadrature checks at 48 versus 80 nodes agreed to within 8×10^{-15} log-density units on evaluated endpoints.

Three candidates were compared: unmirrored quadratic, unmirrored quartic, and mirrored quartic. Selection used participant–scene-balanced held-out log density averaged equally across T1/T2, before judgment fitting. The mirrored quartic was selected. Each candidate was evaluated on scenes excluded from its fit, but those same held-out scenes were used to select the mirrored quartic; its apparent advantage therefore needs confirmation on independent scenes. Wider windows, coarser centers, transformed surfaces, and fitting to this video cohort make this an adaptation rather than an exact reproduction. It does not condition on elapsed time, pursuit velocity, or local video motion. Exported fixation endpoints do not establish saccade-only events.

S2.3 Transition eligibility, control sampling, and joint-reference results

After preprocessing, 9,153 accepted fixation endpoints were screened for transition eligibility. Eligible transitions required an immediately preceding valid fixation, consecutive indices, finite in-video endpoints, and intervals entirely within five seconds. There were 7,598 fitting transitions. Scoring additionally required movement greater than .01 physical-frame units, matching the historical common comparison sample and leaving 7,386 contributions. Smaller valid transitions remained eligible for reference fitting before scene exclusion.

Thirty endpoints were sampled per transition from the fitted joint distribution, retaining video times and fixation durations. Candidates required valid motion interpolation support across the union of lag bins. Observed and control values were duration-aggregated before their excess was divided by the SD of control-replicate trial averages. The sampler used seed 20261009 and transition-specific random streams. Controls were reused across 0-, 100-, and 200-ms scoring lags. Positive lags sampled earlier motion fields. No scoring failures occurred. The joint reference does not match each observed amplitude or endpoint eccentricity and therefore defines a different spatial contrast from the primary reference.

The selected reference improved balanced held-out log density over the historical axiswise reference by .0385 nats per endpoint for T1 and .0956 for T2. That axiswise comparator used independent truncated normals, own-axis linear locations, and constant underlying scales. It is retained as an ablation rather than the featured transition sensitivity. All 5,600 local fits across mirrored and unmirrored adaptations converged without bound hits. Nevertheless, nominal 90% density regions covered 85.6% of T1 and 85.0% of T2 endpoints, and predicted mean amplitudes exceeded observed means. Improved log density did not establish full calibration.

At zero lag with 30 controls, the fitted T2 grand mean was .156 control SDs ($p = .324$). The AI-minus-real contrast was −.388 SD units (95% CI $[-.735, -.040]$, $p = .041$), but its mixed model was singular. This secondary class contrast is specification-sensitive. The judgment interaction was 2.60 ($[1.38, 4.91]$, $p = .0032$); the real-clip slope was OR

= .46 ([.28, .76], p = .0023), and the AI-clip slope was 1.20 ([.80, 1.79], p = .375). The accepted judgment fit used participant/scene intercepts after a singular participant/scene/video attempt. The rescored eccentricity interaction on the same transition sample was 3.07 ([1.51, 6.23]), p = .0019, using all three intercepts.

At 100 and 200 ms, joint-reference interaction ratios were 2.75 ([1.46, 5.19], p = .0017) and 2.68 ([1.43, 5.02], p = .0021). Across the three candidates, zero-lag ratios ranged from 2.29 to 2.72. All candidate and lag estimates are reported in the accompanying summary tables. A numerical control-count sensitivity for the same fitted mirrored reference used 300 nested controls: its zero-lag interaction was 2.47 ([1.32, 4.63], p = .0047). The scene-level paired check of the 300-control run estimated an AI-minus-real motion-coupling difference of −.391 SD (−.788, .006, p = .053); because the interval includes zero, this secondary class contrast remains uncertain.

S2.4 Exploratory history extensions of the joint saccadic-flow reference

The later comparison used the smaller common history-eligible sample defined above. The saved scene-excluded mirrored joint-quartic model supplied the Gaussian base. A launch-only reference added a regularized affine recalibration of its locations; a matched history extension additionally used the immediately preceding displacement. Covariance remained that of the joint base. Task-specific location corrections were fitted excluding the evaluated scene pair with equal participant–scene cell weights; predictor scaling used training statistics only. The primary penalty was .1, with .01 and 1 as secondary shrinkage checks. Results for all three penalty values (.01, .1, and 1) are reported.

All benchmark references sampled 300 motion-supported endpoints per transition, retaining their first 30 for a nested numerical check. Sampling preserved motion-field standardization, duration-first aggregation, and 0/100/200-ms scoring. Every model used the same 6,601 scoring contributions. Coupling was standardized separately within T2 for each reference; ORs therefore represent reference-specific one-SD contrasts. The primary specification for this benchmark was 300 controls and zero lag. These results must not be substituted for the earlier 30-control, 7,386-contribution analysis without identifying the changed sample.

The primary history extension improved balanced endpoint log density over matched launch-only recalibration by .00134 nats in T1 and .00181 in T2. Gains over the uncorrected joint base were larger (.0346 and .0242), indicating that the matched launch recalibration is the informative comparison for the incremental preceding-movement term. The incremental gain from adding preceding displacement was small in this comparison.

S2.5 Frozen-frame DeepGaze III implementation

DeepGaze III combines image content and fixation history in static-image scanpath prediction [Kümmerer et al. 2022]. We used the publicly released implementation at commit `c7db17e2d1d7ea6468ffdee2cfaddf141095dcff` (https://github.com/matthias-k/deepgaze/tree/c7db17e2d1d7ea6468ffdee2cfaddf141095dcff), together with the released v1.1.0 weights and MIT1003 center-bias prior. All parameters were frozen; there was no task-specific training or recalibration on this cohort. This externally trained, content-aware reference differs from the Gaussian references cross-fitted within this cohort in both information inputs and training.

For each evaluated endpoint, the model received an RGB frame at or before the preceding fixation's offset and up to four contiguous preceding fixation locations. The sample required at least two; unavailable older locations were NaN according to the implementation, and invalid history gaps were not bridged. Frames were selected as $\lfloor t_{\text{launch}} f_{\text{source}} \rfloor$ using the recorded average source rate. The saved source index and frame-age audit gave a mean age of 19.1 ms and maximum 41.6 ms, with no future frame under that timing convention. This is a frozen-frame adaptation, with no optical-flow input, pursuit model, or temporal visual integration.

RGB input size was 1184×888, based on the manuscript's aspect-fit geometry, a fixed approximate 65-cm viewing distance, and the model's 35 pixels/degree scale. This geometry is an adaptation assumption rather than a participant-specific distance measurement. History coordinates were transformed to these image pixels. Log densities were normalized and converted to the common physical-frame units for endpoint comparisons. Calibration used unmasked reference draws; motion-coupling controls were additionally conditioned on the existing feature-support masks.

The released weights loaded with strict parameter matching. Inference using cached features and frozen modules agreed with the implementation's standard forward pass to a maximum log-probability difference of 3.82×10^{-6}; predicted probabilities summed to approximately one, and the batching check passed. We used the released implementation without modifying its source. All 40 clips were processed on CPU. Completion checks verified the 6,756 evaluation endpoints, 6,601 control sets, common row identities, finite scores, normalized probabilities, timing bounds, and accepted primary fit diagnostics.

S2.6 Matched benchmark results and calibration

Table S1 reports the zero-lag, 300-control judgment interaction on the common history-eligible sample. All seven accepted judgment models retained participant/scene intercepts and were nonsingular. DeepGaze's real-clip slope was OR = .60 (95% CI [.36, .98], $p = .042$); its AI-clip slope was 1.01 ([.72, 1.43], $p = .951$). Its interaction interval includes one and substantial positive effects. Separate fits do not test whether reference-specific coefficients differ.

Table S1. Common history-eligible benchmark: 385 T2 trials, 300 controls, zero lag, participant/scene intercepts. All p values are nominal. OR ratios compare class-specific AI-response slopes.

Reference	OR ratio [95% CI]	p
Simplified axiswise (ablation)	2.62 [1.41,4.85]	.0022
Joint quartic	2.47 [1.33,4.58]	.0040
Joint + launch recalibration	2.41 [1.30,4.46]	.0050
Joint + two-fixation history	2.47 [1.33,4.60]	.0043
History, weaker shrinkage	2.48 [1.33,4.61]	.0042
History, stronger shrinkage	2.49 [1.34,4.62]	.0039
DeepGaze III + frame	1.69 [.92,3.13]	.0929

DeepGaze's balanced endpoint log density was 1.84995 in T1 and 1.82142 in T2. Its gains over the joint quartic were .01575 and .02762 nats per endpoint; gains over launch recalibration were −.01752 and .00526. Its predictive advantage was therefore mixed rather than uniform across tasks and comparators. The summaries first average within participant–scene cells and then across cells.

Nominal 90% DeepGaze density regions covered 73.9% of T1 and 74.3% of T2 endpoints, with mean density ranks .326 and .330. Gaussian coverage on this same sample was also imperfect, approximately 84–86%. Predicted DeepGaze mean amplitudes were .166 and .163 physical-frame units, versus observed .116 and .123. The model improved average log density over the joint base without establishing calibrated endpoint probabilities. Figure S4 shows the density-rank distributions. Primary zero-lag inference should be distinguished from the secondary timing checks. DeepGaze's 100-ms interaction was 1.37 ([.80, 2.33], $p = .249$), with participant-only intercepts; the 200-ms estimate was 2.05 ([1.03, 4.06], $p = .040$), with participant/video intercepts. The lag analyses keep the launch-frame reference fixed while sampling earlier motion fields; they are not separate video-prediction models. The nested 30-control zero-lag estimate was 1.46

([.85, 2.51], p = .174), with a participant-only intercept. Changed accepted random structures and multiple comparisons limit comparisons across these secondary specifications. The analysis scripts record the fitted formulas and diagnostics.

S2.7 Interpretation and supporting figures

These references define different spatial contrasts. Conditioning on RGB content can incorporate content-related gaze expectations into the control distribution. The present comparison also changes external training, model family, and history representation, so it does not isolate image content or history as the cause of the smaller DeepGaze point estimate. A matched, separately trained no-image ablation was not available; blanking a frame in the full model would not constitute that ablation.

All results are exploratory in the same cohort with unadjusted p values. Intervals condition on fitted reference scores and do not propagate reference-fitting uncertainty or dependence from overlapping training sets. The references share inherited fixation timing, display geometry, and motion fields; none independently validates optical flow or distinguishes pursuit from saccades. The primary eccentricity result remains relative to its specified control. The content-aware comparison limits claims of motion-specific judgment information beyond learned, content-conditioned gaze expectations; it neither proves the absence of an association nor establishes a formal difference in coefficients.

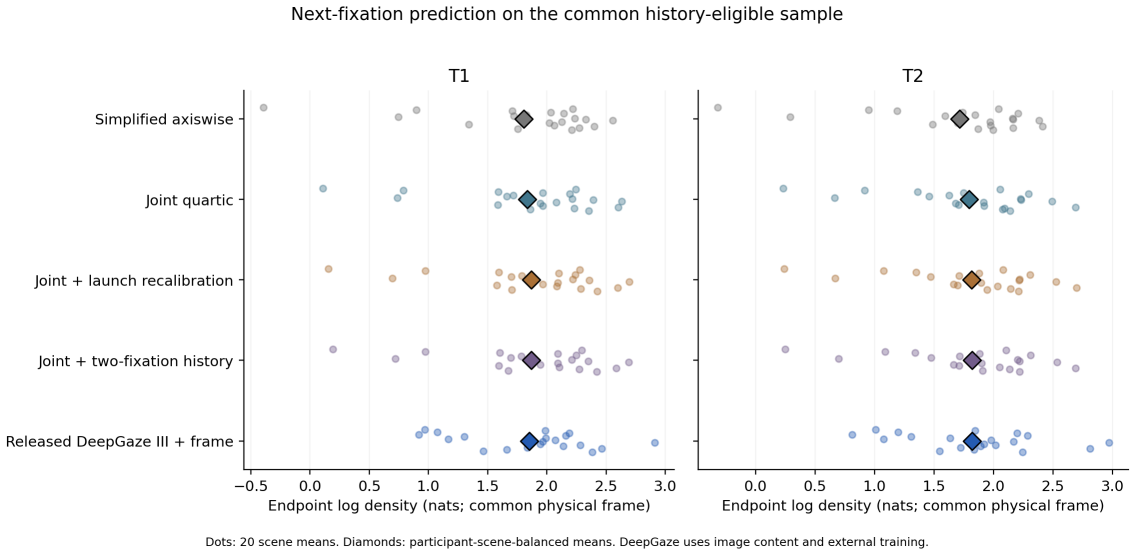


Fig. S1. Endpoint prediction on the common history-eligible sample. Dots show 20 scene means; diamonds show participant–scene-balanced means. DeepGaze uses RGB content and external training. These descriptive comparisons use overlapping data and do not provide independent validation.

S3 Extended judgment models and nested comparisons

S3.1 T2 judgment model

The primary logistic mixed model used maximum-likelihood `glmer` in `lme4` [Bates et al. 2015], with Laplace approximation and the bobyqa optimizer. AI judgment was 1, Real judgment 0, with actual-real as reference. All continuous

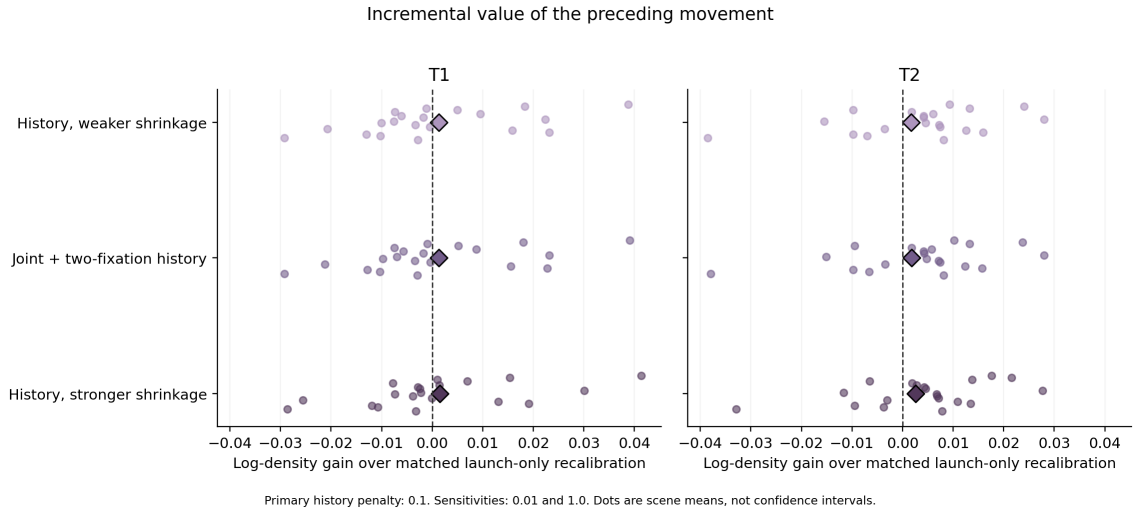


Fig. S2. Incremental endpoint prediction from preceding displacement beyond matched launch recalibration. The primary history penalty is .1; .01 and 1 are secondary checks. Dots show scene means, not confidence intervals. The fitted extension adjusts Gaussian means and does not test all possible history effects.

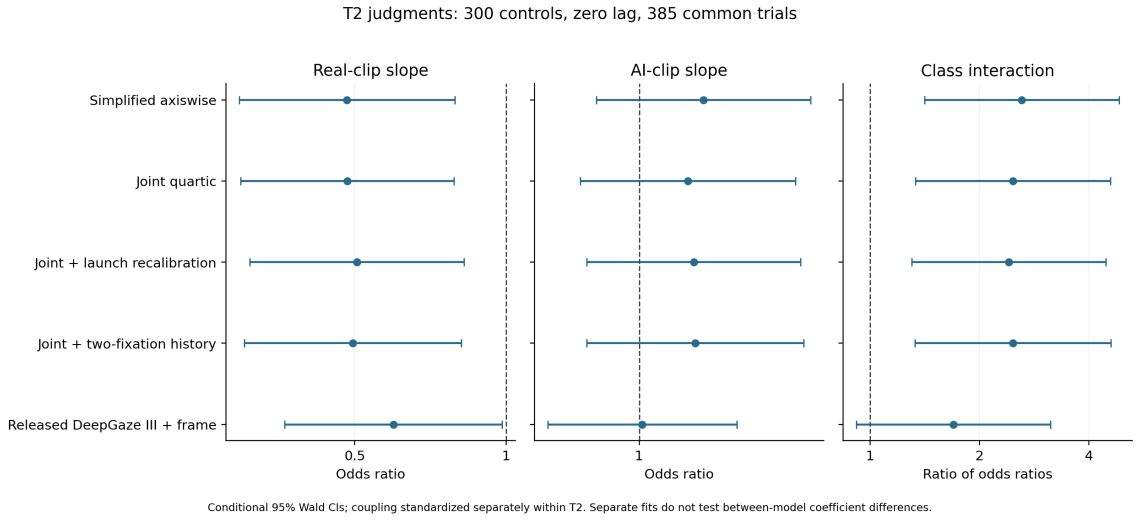


Fig. S3. Matched benchmark judgment slopes and interactions: 385 T2 trials, 300 controls, zero lag. Intervals are conditional 95% Wald intervals. Coupling is standardized separately by reference. Separate fits do not test differences between coefficients.

predictors were standardized across the same 385 T2 trials, using the sample mean and SD. The fixed-effects expression was

$$\operatorname{logit}\Pr(J = \mathrm{AI}) = \beta_0 + \beta_1 z(C) + \beta_2 a + \beta_3 z(C)a + \gamma_1 z(\bar{m}) + \gamma_2 z(N) + \gamma_3 z(D). \tag{S3}$$

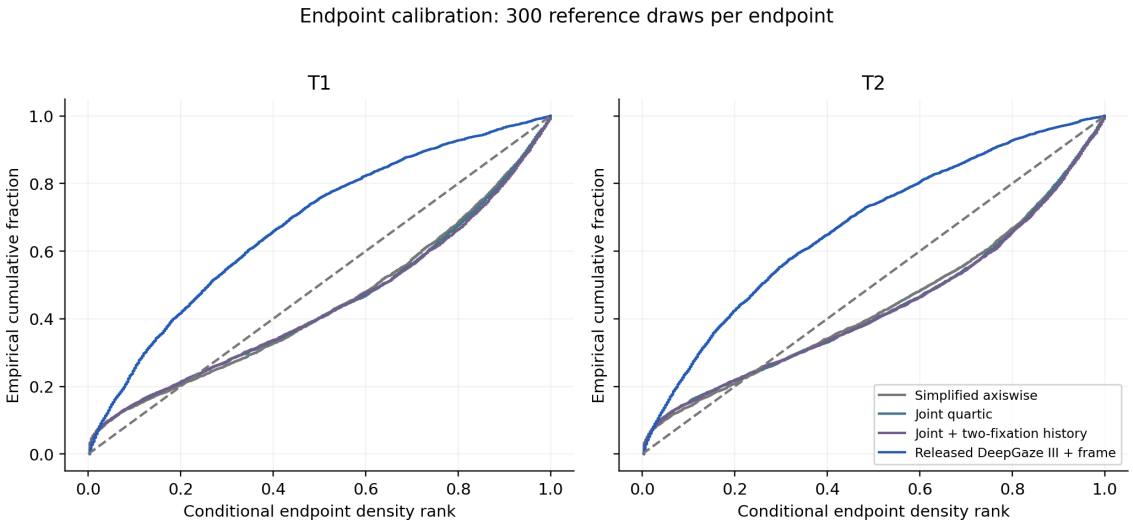


Fig. S4. Conditional endpoint density-rank distributions. The dashed diagonal is the calibrated expectation. The frozen DeepGaze reference shows substantial deviations; the Gaussian references also depart from calibration.

Participant and video had random intercepts. Here C is the primary eccentricity-matched coupling score, $a = 1$ denotes actual AI, $\bar{m}$ mean unscaled motion, N fixation count, and D duration. Conditional coupling slopes are β_1 (real) and $\beta_1 + \beta_3$ (AI), with covariance-based standard errors; $\exp(\beta_3)$ is their ratio of ORs. Two-sided Wald tests and 95% intervals were used. Fits used bobyqa, at most 200,000 function evaluations, and the Laplace approximation (nAGQ=1). No rows were omitted between models.

S3.2 Model comparisons

Every comparison model contained actual class, mean motion, fixation count, and duration, with participant/video random intercepts. *Adjusted coupling* was the primary model above. *Conventional* added scanpath length, dispersion, and entropy to the common adjustments; *Combined* included these descriptors together with coupling and its actual-class interaction. Thus, the primary judgment model itself enters the comparison. The class-specific Conventional model additionally allowed the slopes of scanpath length, dispersion, entropy, count, and duration to differ by actual class; its Combined counterpart added coupling and its class interaction. These two added coupling terms were assessed together.

Let $B = \beta_0 + \beta_a a + \gamma_m z(\bar{m}) + \gamma_N z(N) + \gamma_D z(D)$ denote the common fixed part, $K = \beta_C z(C) + \beta_{aC} a z(C)$ the coupling block, and $G = \gamma_L z(L) + \gamma_S z(S) + \gamma_E z(E)$ the path-length, dispersion, and entropy block. With the same participant/video

random intercepts in each row, the fixed parts were

$$\text{Adjusted coupling}: \quad B + K, \tag{S4}$$

$$\text{Conventional}: \quad B + G, \tag{S5}$$

$$\text{Combined}: \quad B + G + K, \tag{S6}$$

$$\text{Conventional, class-specific}: \quad B + G + a\{\delta_L z(L) + \delta_S z(S) + \delta_E z(E) + \delta_N z(N) + \delta_D z(D)\}, \tag{S7}$$

$$\text{Combined, class-specific}: \quad B + G + a\{\delta_L z(L) + \delta_S z(S) + \delta_E z(E) + \delta_N z(N) + \delta_D z(D)\} + K. \tag{S8}$$

The last two expressions describe separate models, not shared fitted coefficients. Adding K contributes two fixed parameters; adding G to Adjusted coupling contributes three. Conventional and Adjusted coupling are non-nested and were compared by AIC/BIC, not an LR test.

Table S2. Judgment-model fit for the same 385 T2 trials and participant/video intercepts. Parameter counts include fixed coefficients and random-intercept variances.

Model	Parameters	Log likelihood	AIC	BIC
Adjusted coupling	9	−201.985	421.97	457.55
Conventional	10	−207.159	434.32	473.85
Combined	12	−201.950	427.90	475.34
Conventional, class-specific	15	−205.942	441.88	501.18
Combined, class-specific	17	−200.749	435.50	502.70

S3.3 Nested frequentist judgment-model comparisons

All comparisons in Table S3 use identical rows, scaling, and participant/video random intercepts. The coupling-block test concerns its main effect and class interaction jointly; it does not test the interaction alone. Negative ΔAIC favors the larger model on that criterion.

Table S3. Nested additions to the judgment model. ΔAIC is larger minus smaller model.

Smaller → larger	df	LR χ^2	p	ΔAIC
Conventional → Combined	2	10.42	.00546	−6.42
Adjusted coupling → Combined	3	0.07	.995	+5.93
Conventional, class-specific → Combined, class-specific	2	10.39	.00555	−6.39

Table S4. Coupling associations per one T2 SD, from the participant/video models. Values are AI-response ORs with 95% Wald intervals; the final column tests the coupling-by-class interaction.

Model	Actual real	Actual AI	Ratio of ORs	p
Adjusted coupling	.58 [.36,.94]	1.80 [1.08,3.00]	3.12 [1.54,6.33]	.00158
Combined	.58 [.35,.94]	1.80 [1.07,3.01]	3.12 [1.53,6.38]	.00182
Combined, class-specific	.58 [.35,.96]	1.83 [1.09,3.09]	3.16 [1.53,6.52]	.00182

The real- and AI-clip associations had the same directions across these specifications.

S3.4 Random-structure diagnostics and sensitivity

All five participant/video models converged, were nonsingular at tolerance 10^{-4}, retained the same 385 rows, and had finite coefficients and standard errors with positive-definite estimated Hessians. No numerical convergence messages or coefficient/standard-error flags were recorded.

With participant, scene-pair, and video intercepts, the scene-pair variances were effectively zero: 5.98×10^{-14} for Adjusted coupling, zero for Conventional, and 1.56×10^{-9} for Combined. All three fits were singular; Adjusted coupling also had a negative minimum Hessian eigenvalue (−.119). The participant/video models gave the same maximized log-likelihoods to within 3×10^{-8} while retaining individual-video clustering. Their diagnostics support the common participant/video specification. Singularity was recorded separately from optimizer convergence.

Table S5. Random-intercept sensitivity for the three core models. PS denotes participant/scene-pair, PV participant/video, and PSV all three intercepts. The PSV fits are singular.

Model	Intercepts	Log likelihood	AIC	Scene variance
Adjusted coupling	PS	−205.344	428.69	.536
Conventional	PS	−212.590	445.18	.561
Combined	PS	−205.216	434.43	.579
Adjusted coupling	PV	−201.985	421.97	—
Conventional	PV	−207.159	434.32	—
Combined	PV	−201.950	427.90	—
Adjusted coupling	PSV	−201.985	423.97	$< 10^{-12}$
Conventional	PSV	−207.159	436.32	0
Combined	PSV	−201.950	429.90	$< 10^{-8}$

The participant/scene-pair models were nonsingular. Their coupling increment was LR $\chi^2(2) = 14.75$, $p = .000627$, $\Delta\text{AIC} = -10.75$; the class-specific Conventional comparison gave $\chi^2(2) = 15.20$, $p = .000501$, $\Delta\text{AIC} = -11.20$. The three-descriptor increment beyond Adjusted coupling gave $\chi^2(3) = 0.26$, $p = .968$, $\Delta\text{AIC} = +5.74$. The Adjusted coupling interaction ratio was 3.31 ([1.77, 6.19]), compared with 3.23 ([1.71, 6.09]) in Combined and 3.35 ([1.76, 6.38]) in class-specific Combined. No LR test compared models with different random structures.

S3.5 Supporting irregularity and gaze-H_0 additions

The irregularity and gaze-H_0 comparisons used separate participant/scene models, with all 385 trials and T2-wide standardization. Their common baseline contained actual class, mean motion, all five conventional measures, coupling, and its class interaction. These supporting models have the same fixed predictors as Combined, but a different random structure. Adding irregularity and its class interaction gave LR $\chi^2(2) = 1.51$, $p = .471$, $\Delta\text{AIC} = +2.49$.

Adding standardized $\log(1 + P_{H_0}^{\text{gaze}})$ and its class interaction to this baseline gave LR $\chi^2(2) = 1.61$, $p = .446$, $\Delta\text{AIC} = +2.39$. The added model had AIC 436.82 versus 434.43. Its gaze-H_0 AI-response ORs were .77 ([.43, 1.37]) for real clips and .69 ([.38, 1.25]) for AI clips. Adding the same gaze-H_0 block to Conventional gave $\chi^2(2) = 1.79$, $p = .409$, $\Delta\text{AIC} = +2.21$. A participant/video specification for the Combined baseline gave $\chi^2(2) = 1.98$, $p = .372$. These additions do not alter the primary coupling score. The separate Bayesian comparison below also uses participant/scene intercepts.

S3.6 Bayesian gaze-H_0 addition and prior sensitivity

Separately fitted Bernoulli-logit models compared Conventional+Motion with the same model plus two class-specific standardized $\log(1 + P_{H_0}^{\text{gaze}})$ slopes. Both used participant and scene random intercepts and the same 385 rows. Shared

log-odds coefficients had Normal$(0, 0.7)$ priors, the intercept Normal$(0, 1.5)$, and random-effect SDs Exponential(1). The added real and AI slopes had independent, identical Normal$(0, s)$ priors, avoiding the class asymmetry of equal priors on a reference-coded main effect and interaction.

Table S6. Class-symmetric prior sensitivity. BF_{10} is the marginal likelihood of the gaze-H_0-added model divided by baseline; values below one favor baseline. Medians and ranges are over three bridge-sampling repetitions, not sampling-uncertainty intervals.

Added-slope prior SD s	Median BF_{10}	Bridge-run range	BF_{01}
0.25	.679	.673–.683	1.47
0.50 (primary)	.369	.361–.373	2.71
1.00	.134	.134–.135	7.48

At the primary prior, relative support for baseline was modest; strength depended on prior width. Four chains provided 16,000 post-warmup draws per fit; maximum $\widehat{R} = 1.002$, minimum bulk effective sample size 3,113, and no divergences or maximum-tree-depth hits were recorded. These Bayes factors arise from separate fits, not conversion of the frequentist p value, and do not prove absence of a gaze-topology relationship.

S3.7 Actual-AI-only sensitivity

Within 193 AI trials, a participant/scene model used within-AI-standardized coupling plus motion, count, and duration: OR 1.93 ($[1.10, 3.37]$), $p = .0220$. Three-intercept attempts failed to converge. The primary result remains the class-specific all-T2 comparison.

S4 Grouped holdout diagnostics

The grouped diagnostic used five-fold holdouts of whole participants or whole scene pairs (both clip versions), with class-aware L2-penalized logistic regression (scikit-learn inverse regularization strength fixed at 1), training-fold-only scaling, and no ID predictors. Five feature sets were compared: Conventional, Motion, Irregularity (irregularity coupling and its class interaction), Both couplings, and Conventional + both. Coupling models retained the actual-class interaction. AUC and Brier scores were pooled over all out-of-fold probabilities. Actual clip class was supplied in every compared model; these analyses estimate the viewer's response conditional on known class and do not assess detection of unknown class.

Table S7. Descriptive class-aware five-fold holdouts on the same 385 trials. All feature sets include known actual class and mean motion. AUC is pooled ranking performance; lower Brier scores are better.

	Participants		Scene pairs	
Feature set	AUC	Brier	AUC	Brier
Conventional	.703	.203	.687	.208
Motion	.766	.193	.768	.193
Irregularity	.740	.197	.718	.197
Both couplings	.766	.194	.765	.194
Conventional + both	.753	.198	.730	.207

Motion and Both had similar participant-held-out AUCs (.766); Motion had higher scene-held-out AUC (.768 vs. .765) and essentially identical Brier scores in both schemes (Table S7). These within-dataset diagnostics have no formal uncertainty intervals or external validation.

These grouped diagnostics use feature sets distinct from the mixed models in Supplementary Section S3. Here, Motion contains actual class, mean motion, coupling, and its class interaction, without fixation count and duration; Conventional contains all five conventional measures. The fitted models are penalized logistic regressions, not the participant/video mixed models used for the main nested comparisons. They do not validate the transition-reference coupling measures.

S5 Topological construction and video-field diagnostics

S5.1 Cubical persistence of video structure

We computed cubical sublevel persistence on $1 - A$, so cells with higher irregularity values entered the filtration earlier. Voxels unsupported in A were zero-filled; after inversion, they had filtration value 1. We also added a one-voxel outer border at value 1. These cells therefore entered at the final threshold, $\tau = 1$. We used giotto-tda [Tauzin et al. 2021] with coefficients in $\mathbb{Z}_2$ and reduced homology, which omits the essential H_0 class. Let $\mathcal{D}_0$ denote the set of H_0 persistence intervals, with birth b_i and death d_i. Total persistence was the sum of the lifetimes of finite, positive-length intervals:

$$P_{H_0}^{\text{video}} = \sum_{\substack{i \in \mathcal{D}_0 \\ b_i, d_i \in \mathbb{R} \\ d_i > b_i}} (d_i - b_i). \tag{S9}$$

The intervals track connected regions merging across adjacent spatial and temporal cells. Their lifetimes are differences in filtration threshold, not durations in seconds or counts of visible objects. The scalar summarizes topology in the per-video normalized, processed irregularity field; video and gaze fields were normalized separately (Supplementary Section S5.3).

S5.2 Gaze density fields and gaze persistence

Each eligible fixation contributed its within-window duration to overlapping temporal bins, distributed spatially by a Gaussian of SD 2.5 grid bins, truncated at four SDs and renormalized at image boundaries. With overlap Δt_{kt} and normalized kernel K_k, the time-by-x-by-y field was

$$G(t, x, y) = \sum_k \Delta t_{kt} K_k(x, y). \tag{S10}$$

Raw mass summed to eligible fixation duration. Each trial field was divided by its 99th percentile and clipped to $[0, 1]$; the same inversion, padding, reduced persistence, and finite-lifetime sum produced gaze H_0. Zero indicates no retained positive finite intervals. This report concerns the duration-weighted cubical field.

S5.3 Cubical persistence construction

At threshold τ, cells enter when $1 - A$ or $1 - G$ is at most τ. Components are born as separate regions and die on merging with older regions. Diagram points (b_i, d_i) and barcode segments $[b_i, d_i]$ display the same intervals. Total persistence sums all finite positive lifetimes from the complete padded time–space volume, including short intervals. Reduced homology excludes the essential component. Video and gaze fields encode different quantities and are scaled separately.

S5.4 Video-field quality and influence

Omitting the outer border while retaining zero-filled unsupported voxels reproduced the primary H_0 and H_1 totals; zero-valued padding changed all 40 totals. Registration failed for 39/119 Scene-03 AI frame pairs and 117/119 real pairs. Scene-04 AI had mean unscaled motion of 7488.9 derivative pixels/s despite no logged registration failures, with a broad high-valued region requiring flow-level review. Real scenes 04/07/09 each had more than 120 near-static frame pairs; per-video normalization may amplify small residuals. These three scenes accounted for 56.7% of the absolute paired differences; excluding them yielded a mean difference of −69.76 (95% CI [−127.48, −12.04]).

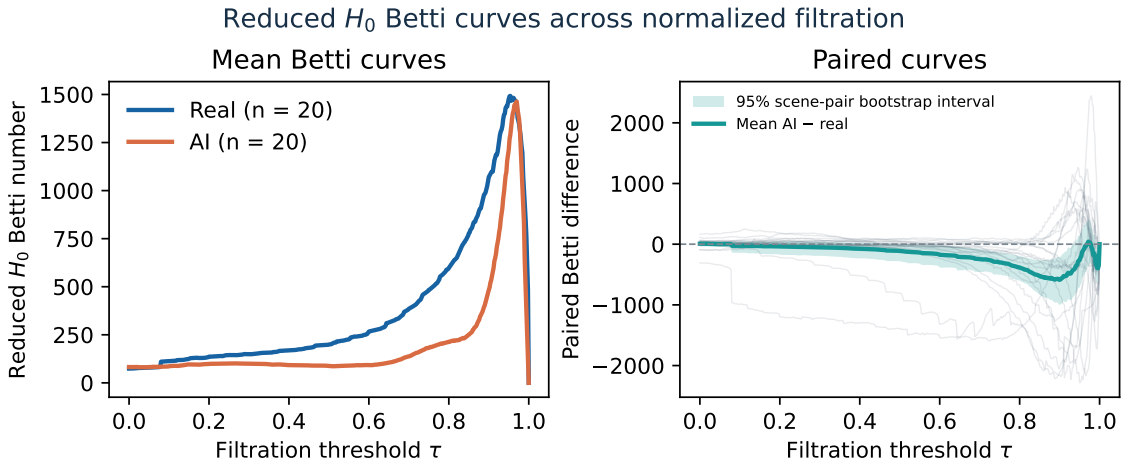


Fig. S5. Reduced H_0 Betti curves across the normalized filtration. Mean class curves and paired AI-minus-real curves decompose the persistence contrast. Their integral summarizes the same total-persistence difference. The band is the plotted scene-pair bootstrap interval; no threshold-wise tests were performed.

S6 Exploratory clip properties and gaze topology

S6.1 Exploratory video change and between-viewer gaze topology

Raw pixel change was the mean absolute difference between consecutive frames of the standardized 24-Hz grayscale derivative, with intensities divided by 255. Frame-wise intensity topology used adjacent-frame averages on the common $60 \times 60 \times 80$ grid. Reduced two-dimensional H_0 diagrams used gray-level sublevel sets; mean 1-Wasserstein distance between consecutive diagrams, computed with giotto-tda [Mileyko et al. 2011; Tauzin et al. 2021] ($p = 1$, approximation tolerance .01), described instability. To compare it with generic change, we regressed log instability on log pixel change across 40 clips and tested the paired AI-minus-real residual contrast.

For each clip, mean pairwise 1-Wasserstein distance between its T2 viewers' cubical gaze-H_0 diagrams described between-viewer topology similarity. Diagram computation used duration-weighted time–space fields with the same Wasserstein settings. Smaller distances mean more similar persistence summaries; they do not retain absolute fixation location or time. The class contrast used 20 paired clip means. Trial typicality was negative mean distance to the other viewers of that clip, excluding the trial itself. Spatial and space–time field correlations were secondary comparisons.

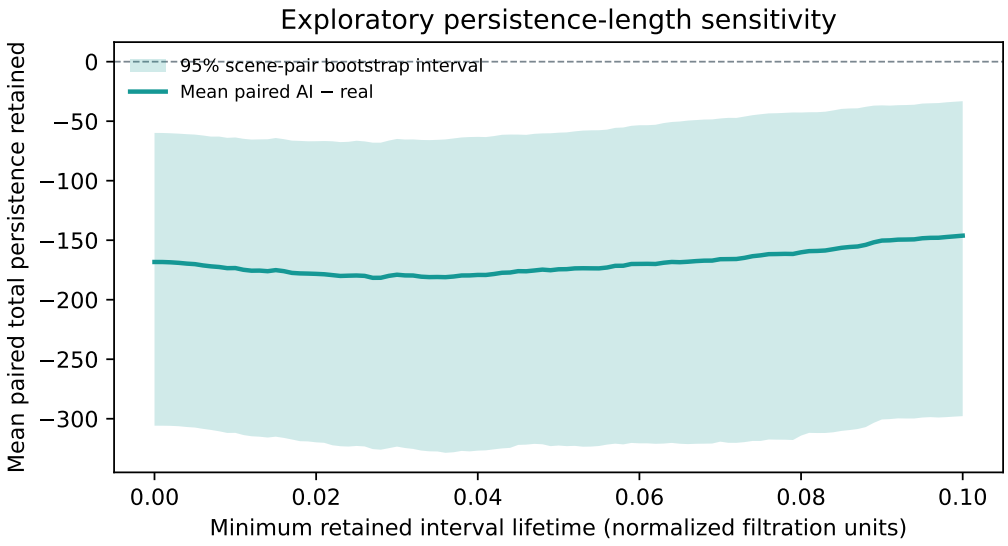


Fig. S6. Exploratory lifetime-cutoff sensitivity of the mean paired video-H_0 difference. The unthresholded analysis is primary; the plotted scene-pair bootstrap band is descriptive and does not constitute independent confirmation.

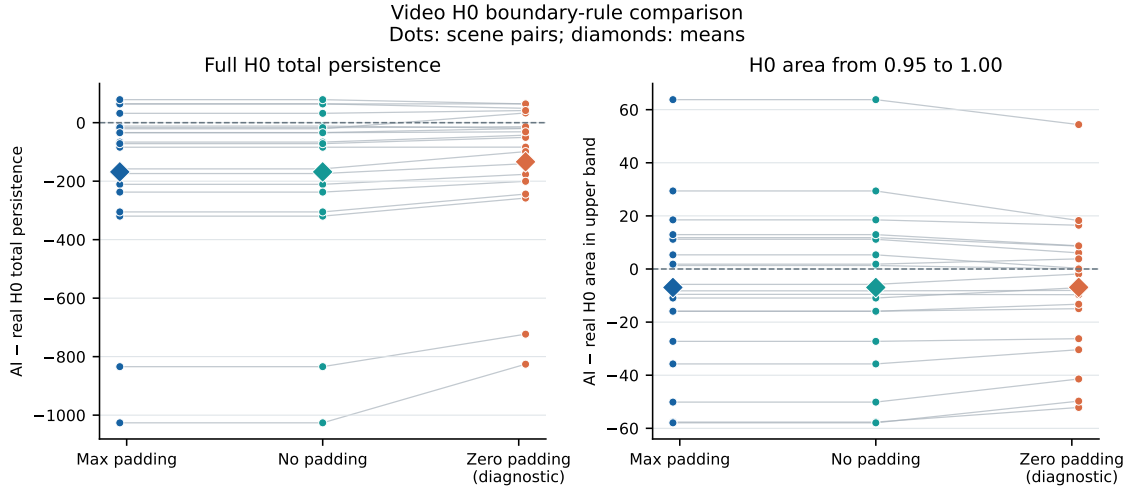


Fig. S7. Video-H_0 boundary-rule diagnostic. Scene-pair differences and means are shown for maximum-value padding, no outer padding, and zero-valued padding. The upper-filtration-band comparison is descriptive. Equality of totals does not establish equality of full diagrams or of gaze analyses.

Pooled-viewer fields averaged raw duration-weighted fields across a clip's T2 viewers, then applied their own 99th-percentile normalization and persistence calculation; pooled H_1 sums finite loop lifetimes.

S6.2 Exploratory irregularity, clip, and joint judgment models

Separate irregularity-coupling judgment fits used actual class, its interaction with standardized coupling, and motion, count, and duration covariates. Continuous variables were scaled within the 385 T2 rows. Irregularity-only and joint motion/irregularity fits used participant and scene intercepts. Incremental LR comparisons retained a common two-intercept structure and identical rows, including Conventional+Motion versus Conventional+Motion+Irregularity.

Clip-property linkage used $\operatorname{logit}\Pr(J = \mathrm{AI}) = \alpha_0 + \alpha_1 a + \alpha_2 z(V) + \alpha_3 a z(V)$ with participant and video intercepts, retaining clip clustering. Here V was log pixel change or log video H_0 for the focal links. A separate quasi-binomial fit used each of 40 clips' AI/Real response counts; its predictor scaling used 40 clip values, whereas GLMM scaling used 385 trial rows. It checked overdispersion without treating repeated responses as independent clip-level observations.

A unified judgment GLMM contained actual class, standardized raw video H_0, standardized $\log(1 + P_{H_0}^{\mathrm{gaze}})$, all interactions, and motion, count, and duration covariates. The nonsingular fit used participant and video intercepts; its reduced model removed only the three-way interaction. A coupling-added sensitivity was fitted separately. These response models differ from the video-to-gaze model, whose outcome is gaze H_0.

S6.3 Exploratory video-to-gaze topology model

The zero-safe linear mixed-model outcome was

$$Y = \log\left(1 + P_{H_0}^{\mathrm{gaze}}\right), \tag{S11}$$

with standardized video H_0, actual and judged class, all interactions, and motion, count, and duration covariates; real was reference for both factors. Continuous predictors were standardized within T2. Participant, scene, and video intercepts gave a nonsingular maximum-likelihood `lmer`/`bobyqa` fit, with Satterthwaite tests [Kuznetsova et al. 2017]. A machine-epsilon $\log(P_{H_0}^{\mathrm{gaze}} + \epsilon_{\mathrm{mach}})$ outcome was retained as a sensitivity comparison on the same 385 trials (Supplementary Section S6.7).

S6.4 Irregularity coupling

Applying the same control scheme to temporal irregularity yielded a weaker class-dependent association: ratio of ORs = 2.08 ([1.17, 3.71]), $p = .0126$. The actual-real AI-response OR was 0.61 ([0.39, 0.95]), $p = .0299$, while the actual-AI OR was 1.26 ([0.87, 1.84]), $p = .220$. With uniform controls the interaction ratio was 0.98 ([0.56, 1.72]), $p = .952$, showing sensitivity to the spatial baseline. In the joint participant/scene fit, the motion ratio remained 2.91 ([1.49, 5.68]), $p = .00178$, whereas the irregularity ratio was 1.44 ([0.78, 2.67]), $p = .243$.

Same-structure comparisons gave no irregularity increment: adding its slope and class interaction to Conventional+Motion gave LR $\chi^2(2) = 1.51$, $p = .471$, $\Delta\mathrm{AIC} = +2.49$; adding them to the simpler Motion model gave $\chi^2(2) = 1.37$, $p = .505$, $\Delta\mathrm{AIC} = +2.63$ (Supplementary Section S3.3).

S6.5 Video properties and clip-level judgments

Across 20 matched scenes, mean video H_0 total persistence was 206.48 for AI and 374.78 for real clips. The paired AI-minus-real difference was −168.30 (95% CI [−302.81, −33.80]), $t(19) = -2.62$, $p = .0169$. The field describes normalized flow/photometric inconsistency. Scenes 04/07/09 supplied 56.7% of absolute differences; unresolved flow/registration checks and per-clip scaling limit any physical or semantic interpretation (Supplementary Section S5.4).

Raw pixel change was higher in 18 of 20 AI clips, with mean paired difference .00369 in normalized gray-level units ([.00192, .00546]), $p = .000333$. Within actual-AI clips, one SD higher *log* pixel change was associated with more AI responses, OR 2.62 ([1.31, 5.23]), $p = .00650$; the actual-real OR was 0.65 ([0.31, 1.35]), $p = .247$. The class interaction ratio was 4.03 ([1.44, 11.25]), $p = .00785$. The 40-clip quasi-binomial check retained the AI slope's direction (OR 2.20, $p = .0167$), with its own clip-level standardization (Figure S8A).

The log video-H_0 judgment link was less clear: the AI slope was OR 0.43 ([0.18, 0.99]), $p = .0483$, but the clip-level check gave $p = .0767$. Intensity-frame topological instability was higher in all 20 AI clips ($\Delta = .232$, [.131, .332]; $p = .000118$). After residualizing log instability on log pixel change, no clear paired residual contrast remained ($\Delta = .069$, [−.105, .243]; $p = .416$). Thus generic frame change is an important comparison for this exploratory topology signal.

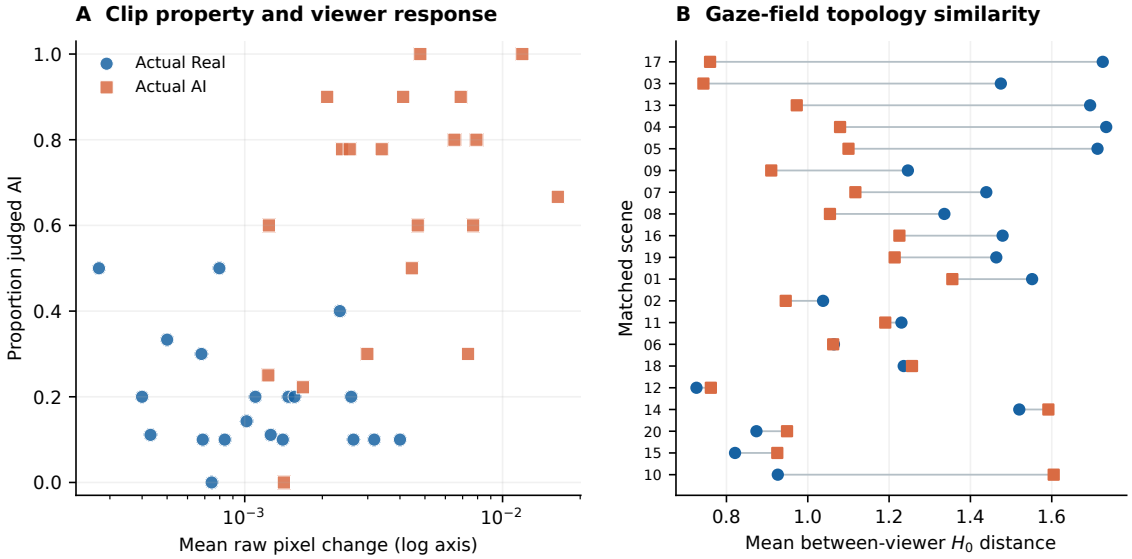


Fig. S8. **Two clip-level summaries. A**, Each point is one of 40 clips: mean raw pixel change versus its proportion of AI judgments. Colors and shapes show *actual* class; the vertical quantity is the viewers' response. **B**, Lines join mean between-viewer gaze-H_0 diagram distances for real and AI clips of each of 20 matched scenes. Smaller distances indicate more similar topological summaries. The mean paired AI-minus-real difference was −0.224 (95% CI [−0.402, −0.046]).

S6.6 Gaze topology: viewer agreement and judgment information

Between-viewer gaze-H_0 diagram distances were lower for AI clips. The mean paired difference was −0.224 (95% CI [−0.402, −0.046]), $t(19) = -2.63$, $p = .0164$ (Figure S8B). Trial typicality differed by actual class (+0.50 SD, $p = .0135$), but its judgment slopes were imprecise: actual-real OR 1.22 ([0.80, 1.86]) and actual-AI OR 1.06 ([0.69, 1.64]), interaction $p = .644$. All eleven trial-topology class interactions had $p > .17$. Thus the clip-level similarity difference did not translate into a clear individual-response association.

With a zero-safe gaze-H_0 *outcome*, video H_0 covaried differently across actual-class/judgment cells: interaction $\beta = .320$ (SE .157), $p = .0424$. The machine-epsilon outcome gave $\beta = 4.624$ (SE 1.657), $p = .00557$, mapping three valid zeros to approximately −36. Both fits used the same 385 trials; their different scales make this a sensitivity result (Supplementary Section S6.7). In the unified *judgment* GLMM, actual class × video H_0 × gaze H_0 was imprecise,

$\beta = .204$ (SE .828), $p = .805$; removing that term gave LR $\chi^2(1) = .061$, $p = .804$. Separately, adding trial gaze H_0 and its class interaction after Conventional+Motion gave $\chi^2(2) = 1.61$, $p = .446$, ΔAIC = +2.39. The Bayesian comparison is in Supplementary Section S3.6. Selected pooled-viewer H_1 links appear in Supplementary Section S6.8; those clip summaries use the same viewers whose judgments are modeled.

S6.7 Video-to-gaze association and zero handling

The outcome here is gaze H_0, rather than judgment. The same three-intercept model retained all 385 T2 trials, including three actual-AI/judged-AI zero-H_0 trials with 7/9/9 fixations and 4.87–4.91 s coverage. The zero-safe $\log(1 + P_{H_0}^{\text{gaze}})$ interaction was $\beta = .320$ (SE .157, Satterthwaite df 366.4), $p = .0424$; the machine-epsilon log outcome gave $\beta = 4.624$ (SE 1.657), $p = .00557$. The epsilon scale maps zeros to approximately −36. Its actual-AI/judged-AI video-H_0 slope was 4.247 ([1.571, 6.923]). Outcome-scale dependence prevents direct comparison of coefficient magnitudes. Neither the unified model nor the coupling-added sensitivity model supported a video-H_0 by gaze-H_0 by actual-class interaction ($p = .805$ and $p = .965$, respectively).

S6.8 Selected exploratory extensions

Mean irregularity coupling was +1.56 control SDs (SE .19); its actual-class contrast was not supported ($p = .409$). Core regions joined the five longest-lived video-H_0 components at each midpoint threshold, $b_i + .5(d_i - b_i)$, using 26-neighbor connectivity. Their dwell excess used the same motion controls and was measured in fixation-time percentage points because sparse regions produced unstable control SDs. Top-five core excess was +1.2 points, $p = .041$, with no increment beyond motion, LR $\chi^2(2) = 1.16$, $p = .560$.

Table S8. Selected secondary clip-level gaze-H_1 links. ORs are AI-response slopes within actual-AI clips per SD of the raw feature across 385 rows, from participant/video GLMMs. The separate quasi-binomial check scales 40 clip values and uses their response counts. All tests are nominal.

Clip feature	OR [95% CI]	GLMM p	Quasi p
Mean between-viewer H_1 distance	.52 [.30, .90]	.0184	.0290
Pooled-viewer gaze-field H_1	.51 [.31, .86]	.0108	.0195

Neither class interaction was supported ($p = .284$ and .317). These clip summaries use the same viewers whose responses are used to fit the linkage. Motion-field topology and periodicity showed no class contrast. Mapper, a cover-and-cluster summary guided by a low-dimensional lens [Singh et al. 2007], used standardized T2-trial features, the first principal component as its lens, 10 intervals with 40% overlap, and single-linkage clustering at the 10th percentile of pairwise feature distances; nodes required at least three trials. It produced a 10-node chain. Node judgment-rate heterogeneity was not distinguishable under 2,000 label permutations ($p = .078$). This graph was descriptive and was not used as judgment evidence.

S7 Verbal reasons, coding checks, and questionnaire recount

S7.1 Verbal reasons: corpus and coding

We examined experimenter-transcribed spoken explanations recorded after T2 judgments. The full source corpus contained 381 nonempty reports: 167 from physics set S1 and 214 from S2, with median length six words. Almost all

accompanied AI responses (380 of 381). S2 was used only for descriptive coding and participant-summary checks in Supplementary Section S7.4; all reason-profile and reason–gaze models reported here used eligible S1 T2 trials.

Reports were linked to reconciled responses using participant, project, task, stimulus set, and media name; response labels agreed on all 385 eligible S1 judgment trials. Of 161 AI responses, 160 had a report: 122 hits on AI clips and 38 false alarms on real clips, contributed by 40 participants viewing 38 clips. One AI response lacked a reason. The one eligible report accompanying a Real response was excluded. Conditioning on AI responses makes hit versus false alarm identical to actual-AI versus actual-real class; the comparison cannot isolate correctness from stimulus properties.

Four overlapping binary content labels distinguished movement and temporal behavior (*dynamics*), physical or causal implausibility (*physical event*), changes in an object's presence or properties (*object change*), and static visual/rendering properties (*appearance*). Two further labels recorded a named object, material, or region (*localized*) and explicit uncertainty (*uncertain*). A report with any content label was a *concrete-cue* report; its complement was *impression-only*. A composite combined dynamics and physical-event reports. Labels describe stated information and do not verify a cue's presence in the clip.

The two language-model coders were Claude Opus 5.5 and GPT-6 Sol (October, 2026). A human reviewer who was not involved in the gaze analysis provided both coders with the same instructions and practice sheets for 20 reports, then gave feedback on their practice coding. The feedback clarified that evaluative wording alone did not justify assigning a physical-event label. The coders independently applied the scheme to shuffled, text-only sheets with anonymous reason IDs, without participant or video identities, actual class, correctness labels, or gaze data. The first coder's labels were checked against the practice-feedback clarification before the second coder's main labels were inspected. We used both models through their chat interfaces and retained the instructions, practice sheets, and returned label files. Agreement was calculated before reconciliation on 337 main reports, excluding the 24 main-sheet reports quoted as examples in the instructions. A sensitivity check included all 361 main reports. Cohen's κ [Cohen 1960] is reported separately for each label.

The same reviewer then labeled all six indicators on the 29 reports with at least one coder disagreement. The reports were reshuffled under new IDs, with the previous labels concealed from the reviewer. The reviewer's labels replaced every label on those reports, including labels on which the coders had agreed; cells left unmarked after the reviewer completed the positive labels were recorded as zero. These reconciliation counts refer to the full 381-report coding file, comprising the 361 main reports and 20 practice reports: 29 reports received reviewer adjudication, and the other 352 retained the labels on which the coders agreed. Because review was limited to reports with a disagreement, it adjudicated those reports but does not provide a representative estimate of human–model agreement. Label-level checks are given in Supplementary Section S7.4.

S7.2 Exploratory reason profiles and reason–gaze associations

Each binary reason indicator was modeled separately by logistic regression with actual class and a participant intercept. ORs compare hits with false alarms. Localization used ordinary logistic regression after the mixed model failed to converge. These profile models had no video intercept, limiting control for clip differences.

Reason–gaze models used the 160 AI-response trials. Eight indicators were paired with eight gaze outcomes: motion coupling, irregularity coupling, dispersion, entropy, scanpath length, fixation count, log-transformed gaze-field H_0, and spatial revisit fraction. The concrete-cue indicator was omitted because it duplicated the complementary impression-only contrast. Spatial revisit fraction was the proportion of spatial-H_0 merge distances below one visual degree, indexing fixation clustering rather than chronological returns to a named object. At the 90-ppi display and 65-cm

midpoint distance, one degree is approximately 40.2 display pixels (27.6 source-video pixels); the stated 60–70-cm range corresponds to about 37.1–43.3 display pixels (25.5–29.7 source-video pixels). Each gaze outcome was standardized within the 160 trials. Linear models contained the reason indicator and actual class, with crossed participant and video intercepts attempted first, then a participant intercept if needed. A singular participant-only fit was retained and flagged when neither structure was nonsingular. Maximum-likelihood fits used Satterthwaite tests.

The 64 tests included five focal exploratory associations: dynamics with motion coupling, dynamics or physical event with motion coupling, object change with irregularity coupling, and localization with dispersion and spatial revisit fraction.

Two additional models tested actual-class interactions with the two motion-related indicators. All motion-coupling fits retained singular participant intercepts with effectively zero variance and no video intercept; other additive gaze models retained participant and video intercepts. Participant-level comparisons pooling S1 and S2 reports are described in Supplementary Section S7.4. All reason analyses were exploratory and their p values unadjusted.

S7.3 Verbal reasons accompanying AI responses

Concrete cues and stimulus class. Concrete cues appeared in 98 of 122 hit reports (80.3%) and 22 of 38 false-alarm reports (57.9%): participant-intercept OR 4.98 ([1.72, 14.42]), $p = .003$ (Table S9). Impression-only reports were correspondingly more frequent among false alarms (42.1% versus 19.7%); this is the complementary contrast. Dynamics or physical-event cues occurred in 58.2% of hits and 39.5% of false alarms, OR 2.24 ([1.02, 4.90]), $p = .044$. The individual content labels had intervals including one, and appearance and localization showed little separation. Within this AI-response sample, correctness and actual class coincide. The results describe different reported cue profiles for AI and real clips and cannot attribute that difference specifically to successful detection.

Table S9. Reason indicators on 160 S1 AI-response trials. ORs compare hits (actual AI) with false alarms (actual real); labels overlap. Tests are nominal.

Indicator	Hits n/122 (%)	False alarms n/38 (%)	OR [95% CI]	p
Any concrete cue	98 (80.3)	22 (57.9)	4.98 [1.72, 14.42]	.003
Impression-only	24 (19.7)	16 (42.1)	0.20 [0.07, 0.58]	.003
Dynamics or physical event	71 (58.2)	15 (39.5)	2.24 [1.02, 4.90]	.044
Dynamics	48 (39.3)	10 (26.3)	2.08 [0.86, 5.07]	.106
Physical event	34 (27.9)	5 (13.2)	2.67 [0.92, 7.74]	.071
Object change	24 (19.7)	3 (7.9)	3.16 [0.83, 12.01]	.092
Appearance	13 (10.7)	4 (10.5)	1.20 [0.33, 4.34]	.777
Localized	59 (48.4)	20 (52.6)	0.84 [0.41, 1.75]	.646
Uncertain	4 (3.3)	2 (5.3)	0.45 [0.04, 4.49]	.494

Concrete cue and impression-only are complementary. All profile models use participant intercepts except localization, which uses ordinary logistic regression. None includes a video intercept.

Reported cues and gaze sampling. All five focal associations had intervals including zero (Table S10). Dynamics reports did not accompany higher motion coupling: $b = -0.12$ SD ([−0.45, 0.20]), $p = .455$. The dynamics/physical-event composite gave $b = -0.10$ ([−0.42, 0.21]), $p = .512$. Object-change reports showed no clear irregularity-coupling association; localized reports showed no clear dispersion or spatial-revisit association. Actual-class interactions were also inconclusive ($p = .826$ for dynamics; .805 for the composite). These motion-coupling fits retained a singular

participant-only intercept and no video intercept. The findings supply no clear evidence of correspondence for these focal categories, while their intervals and clustering limitations prevent equivalence or dissociation claims.

Table S10. Focal exploratory reason–gaze associations, adjusted for actual stimulus class. Coefficients compare reports with versus without each indicator in SD units within the 160-trial sample.

Reason indicator → gaze outcome	Difference [95% CI]	p
Dynamics → motion coupling	−0.12 [−0.45, 0.20]	.455
Dynamics/physical event → motion coupling	−0.10 [−0.42, 0.21]	.512
Object change → irregularity coupling	−0.10 [−0.49, 0.30]	.631
Localized → dispersion	0.01 [−0.17, 0.19]	.901
Localized → spatial revisit fraction	−0.04 [−0.33, 0.24]	.759

The two motion fits retain singular participant intercepts with effectively zero variance and no video intercept. The other three retain participant/video intercepts. All p values are nominal.

Four of the 59 other reason–gaze tests reached $p < .05$: appearance with higher motion coupling ($b = .57$ SD, $[.07, 1.06]$, $p = .025$), physical events with greater spatial revisit fraction ($b = .36$, $[.04, .67]$, $p = .027$), impression-only reports with fewer fixations ($b = -.38$, $[-.72, -.03]$, $p = .031$), and dynamics/physical-event reports with more fixations ($b = .32$, $[.02, .62]$, $p = .038$). These are exploratory leads among 64 related tests; the appearance–motion fit also has the singular clustering limitation. Coding agreement and participant-summary checks are retained in Supplementary Section S7.4.

S7.4 Verbal-reason coding checks and participant summaries

Pre-reconciliation agreement on 337 reports excluding practice and instruction examples was $\kappa = .926$ for dynamics, .893 for physical events, .927 for object changes, .965 for appearance, .958 for localization, and 1.000 for uncertainty. Percentage agreement ranged 96.7%–100%. Across the six labels, Cohen's κ ranged from .900 to 1.000 when all 361 main-sheet reports were included. These statistics measure agreement between the two language-model labelings rather than accuracy against independent ground truth.

The human review covered 174 assignments on 29 disputed reports. In 36 initially discrepant cells, human labels matched the first coder on 25 and the second on 11. Across all six labels on those reports, the reviewer changed 46 assignments relative to the prior model-agreed version. Human labels replaced the initially shared labels, as well as the discrepancies. Disagreement-based selection prevents estimating human–model reliability for the full corpus.

For ancillary participant summaries, S1 and S2 reasons were pooled to calculate each participant's proportions with a concrete cue, a dynamics/physical-event cue, and a localized reference. Two-sided Mann–Whitney tests compared the shared Strategies' Logic/Intuition and General/Specific labels. The data coded 24 Logic and 16 Intuition participants.

Rank-based AUC was $U/(n_1 n_2)$, describing group separation.

Forty participants contributed 4–19 reports each (median nine). Logic participants ($n = 24$) had a higher mean dynamics/physical-event share than Intuition participants ($n = 16$), .497 versus .377, with AUC .669, $p = .075$. Concrete-cue shares were .727 versus .678, AUC .534, $p = .730$. Specific participants ($n = 16$) had more localized reports than General participants ($n = 24$), .570 versus .471, AUC .638, $p = .146$. These comparisons offered limited correspondence with the earlier classifications; variable numbers of reports from two stimulus sets do not establish stable participant traits. Appearance labels were more frequent in S2 (66/214, 30.8%) than S1 (18/167, 10.8%), across different stimulus sets.

All eight additive motion-coupling reason fits and both interaction fits retained singular participant-only intercepts; the additive participant variances ranged from zero to 1.19×10^{-14}. The other 56 additive gaze fits retained nonsingular participant/video intercepts. No equivalence or mediation tests were conducted.

S7.5 Questionnaire cue recount

The recount used only participants' free-text responses to the participant-level strategy question for each of 40 unique participants. Case-insensitive whole-word matching counted *movement*, *movements*, *motion*, *motions*, *move*, and *moves* as explicit movement/motion terms. A separate physical/causal pattern counted *physics*, *physical*, *unphysical*, *momentum*, and the phrases *cause and effect* and *conservation of mass*. Each participant contributed at most once per category; categories could overlap. Nineteen answers (47.5%) matched movement/motion and 18 (45.0%) matched physical/causal terms. This narrow lexical definition includes the phrase "physical representation" and does not equate every physical-term match with a reported physical-law violation. It does not infer cues from generic realism, smoothness, logic, or implied movement without a listed term. Counts describe stated questionnaire content, not validated strategy prevalence or per-trial behavior. A companion script preserves patterns, matches, and the source hash.

S8 Software and reproduction

We used Python 3.12.7 and R 4.4.2, with `lme4` [Bates et al. 2015], `lmerTest` [Kuznetsova et al. 2017], `brms`/`cmdstanr` and Stan [Bürkner 2021; Carpenter et al. 2017], bridgesampling [Gronau et al. 2020], and giotto-tda for cubical persistence and Wasserstein distances [Mileyko et al. 2011; Tauzin et al. 2021]. The accompanying scripts specify package versions, configuration, commands, and input checksums. The fixed configuration included geometry, five-second window, grid, flow and persistence parameters, location controls, and seed 20260808.

The joint-reference sensitivity used the saved fits, control locations, and lag-specific outputs. Its seed (20261009) differs from the primary coupling seed (20260808). The common-sample benchmark used inputs and random-number streams seeded separately for each reference. DeepGaze inference ran under PyTorch 2.4.0+cpu and torchvision 0.19.0+cpu; its judgment-model diagnostics used R 4.5.1 on Linux. The primary eccentricity-reference models and nested comparisons used R 4.4.2 with `lme4` 1.1.37. The scripts and output tables report software versions, fitted formulas, input checksums, and completion checks.

De-identified derived data and analysis code are available through the anonymized Figshare link: Figshare. Original participant-level fixation exports and the pretrained DeepGaze III checkpoint are not included; the supplied script retrieves the public DeepGaze files for rerunning inference.

References

Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. 2015. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software* 67, 1 (2015), 1–48. doi:10.18637/jss.v067.i01

Paul-Christian Bürkner. 2021. Bayesian Item Response Modeling in R with brms and Stan. *Journal of Statistical Software* 100, 5 (2021), 1–54. doi:10.18637/jss.v100.i05

Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. Stan: A Probabilistic Programming Language. *Journal of Statistical Software* 76, 1 (2017), 1–32. doi:10.18637/jss.v076.i01

Alasdair D. F. Clarke, Matthew J. Stainer, Benjamin W. Tatler, and Amelia R. Hunt. 2017. The saccadic flow baseline: Accounting for image-independent biases in fixation behavior. *Journal of Vision* 17, 11 (2017), 12, 1–19. doi:10.1167/17.11.12

Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46. doi:10.1177/001316446002000104

Gunnar Farnebäck. 2003. Two-Frame Motion Estimation Based on Polynomial Expansion. In *Image Analysis (Lecture Notes in Computer Science, Vol. 2749)*. Springer, 363–370. doi:10.1007/3-540-45103-X_50

Martin A. Fischler and Robert C. Bolles. 1981. Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography. *Commun. ACM* 24, 6 (1981), 381–395. doi:10.1145/358669.358692

Quentin F. Gronau, Henrik Singmann, and Eric-Jan Wagenmakers. 2020. bridgesampling: An R Package for Estimating Normalizing Constants. *Journal of Statistical Software* 92, 10 (2020), 1–29. doi:10.18637/jss.v092.i10

Leslie Kish. 1965. *Survey Sampling*. John Wiley Sons, New York.

Matthias Kümmerer, Matthias Bethge, and Thomas S. A. Wallis. 2022. DeepGaze III: Modeling free-viewing human scanpaths with deep learning. *Journal of Vision* 22, 5 (2022), 7, 1–27. doi:10.1167/jov.22.5.7

Alexandra Kuznetsova, Per B. Brockhoff, and Rune H. B. Christensen. 2017. lmerTest Package: Tests in Linear Mixed Effects Models. *Journal of Statistical Software* 82, 13 (2017), 1–26. doi:10.18637/jss.v082.i13

Yuriy Mileyko, Sayan Mukherjee, and John Harer. 2011. Probability Measures on the Space of Persistence Diagrams. *Inverse Problems* 27, 12 (2011), 124007. doi:10.1088/0266-5611/27/12/124007

Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. 2011. ORB: An Efficient Alternative to SIFT or SURF. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2564–2571. doi:10.1109/ICCV.2011.6126544

Danqing Shi, Lan Jiang, Katherine M. Collins, Shangzhe Wu, Ayush Tewari, and Miri Zilka. 2026. Perceived authenticity drives gaze behavior when watching AI-generated videos of physical scenes. *Scientific Reports* 16 (2026), 25792. doi:10.1038/s41598-026-55437-5

Gurjeet Singh, Facundo M'emoli, and Gunnar Carlsson. 2007. Topological Methods for the Analysis of High Dimensional Data Sets and 3D Object Recognition. In *Proceedings of the Symposium on Point Based Graphics*. The Eurographics Association, 91–100. doi:10.2312/SPBG/SPBG07/091-100

Guillaume Tauzin, Umberto Lupo, Lewis Tunstall, Julian Burella Pérez, Matteo Caorsi, Anibal M. Medina-Mardones, Alberto Dassatti, and Kathryn Hess. 2021. giotto-tda: A topological data analysis toolkit for machine learning and data exploration. *Journal of Machine Learning Research* 22, 39 (2021), 1–6.